

The Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Curated Corpora

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Abstract

When a language model answers from a curated corpus via graph-based retrieval, reported grounding uplift mixes *delivery* of facts already shown with *reasoning* over injected structure. We propose *exposure accounting*: classify every gold item by whether the context exposed it and whether the answer recovered it, and report the four counts. The *copy ceiling*, the recall a verbatim copy of the context achieves, is that table’s scalar and the signed *gain over copy* its net, both judge-free. Across ten models, unaided recall averages 0.26 and grounded recall 0.92, yet gain over copy is uniformly negative (-0.067 to -0.022). Of 11,360 gold-item observations (1,136 target instances under ten models, not 11,360 independent facts), three unexposed items are recovered. A stratified audit of 423 of those observations, under a cross-family judge and a symmetric quote gate, puts the precision of the matcher’s credit at 0.971 [0.941, 0.986]. On 60 unexposed uncredited items the judge finds no hidden correct answer (upper bound 6%); all three unexposed credited items, a census, fail adjudication, so credit beyond exposure is rare and here spurious. Correcting the exposed-item rate for judged relational correctness gives an estimated 0.905 against 0.931 under this judge. On gold targets the scaffold did not expose, lexical recovery falls from 0.121 bare to 0.004 grounded; relational adjudication validates 71 of the 92 bare-arm credits and none of the three grounded-arm credits. Rephrasing out of the graph’s title vocabulary collapses exposure from 0.964 to 0.328, while the absence-keyed fallback fires on 2 of 506. A paired production study lifts judged quality by +0.27 pooled. Negative controls show the served path beating no context, but not that the scaffold’s content matters beyond a well-formed on-corpus block. Evaluations whose gold derives from the injected corpus should report the accounting.

1 Introduction

Enterprises need language models to answer from curated, organisation-controlled corpora that pretraining does not reliably cover. Retrieval-augmented generation bridges parametric memory and required knowledge [15, 29], particularly for long-tail and proprietary facts [36]. Structured, graph-based retrieval further sharpens this approach [3, 9, 19, 42]. Yet when gold answers derive from the injected corpus, headline “grounding uplift” conflates *delivery* of exposed

facts with *reasoning* over injected structure. Published knowledge-graph grounding reports lack a standing control that separates them.

We introduce *exposure accounting*, and the *copy ceiling*¹, as its per-item scalar, together with the signed *gain over copy*. Exposure accounting classifies each gold item by whether the shown text exposed it and whether the answer recovered it, and reports the four resulting counts. The copy ceiling is the recall a no-op extractor of exactly the shown text would achieve, which is the exposed fraction; gain over copy is grounded recall minus that ceiling, which pooled over items is the net of two of the four counts. The counts depend only on a surface matcher applied to fixed text, so they carry the findings; the scalar summarises them. This deterministic, judge-free control is paired with a seed-disjoint placebo, and §10.2 measures what the determinism costs by auditing the matcher against judged assertions on a stratified sample of the sweep. It generalises the discrete input-only baselines used to interpret reading comprehension [25] and natural-language inference [20, 46]. Grounding evaluations whose gold derives from the injected corpus should report the accounting as standard, with the ceiling as its summary [27, 38].

Without assistance, the evaluated models score 0.265 on average (range 0.151–0.375) on in-domain questions and ≈ 0.92 with curated context. Across ten models from eight developer families, gain over copy is uniformly negative (-0.067 to -0.022), while beyond-exposure recovery totals three items in 11,360 item×model observations (the same 1,136 target instances scored under each of ten models, §10.3). Models therefore differ almost entirely in how much surfaced gold they drop. A clarification on “private”: the corpus is not secret. It is published on GitHub and at a public site (§3), so the non-trivial raw scores (0.151–0.375 across models) reflect partial pretraining exposure rather than a genuinely unseen domain; what makes it a useful test bed is that it is organisation-controlled and curated, not that it is hidden. Throughout, “private” denotes the *deployment* pattern (LAN-confined, no cloud egress), not corpus secrecy. Delivery of an authoritative, human-curated source is the product. Throughout, “delivery” is scored as recall of exposed gold target names and nothing more. Relational cor-

¹The term *copy ceiling* is used independently and in an unrelated sense by Liu [33] (a decoding-layer editing mechanism, where it denotes token reachability under a line-level copy primitive; that work is withdrawn), noted here only to head off a terminology collision.

rectness is measured separately, on a stratified sample of the sweep under model adjudication (§10.2); precision of the full answer against fabrication, and the accuracy of the attribution metadata, remain unmeasured here (§17).

How the control came to be built. We built the serving node first and deployed it, and the conventional measurement it produced was a large uplift over the same model answering unaided, of the kind this literature routinely reports. The question we then put to our own result was narrower than that headline: what would a verbatim copy of the text we had already shown the model score on the same questions? That baseline sat above the grounded system on every model we swept, which changed what the uplift could be taken to support. The protocol below is that same check written down, so that another team can run it against its own deployment before reporting an uplift figure.

Three methods below are corpus-general and separable from the node they are measured on: the copy-ceiling computation and signed gain over copy (§5, §7); the seed-disjoint placebo construction (§5.4, §10.7); and the blind before/after judged gate for write paths (§12.1). A fourth, the stratified audit of §10.2, is specific to a lexical matcher but transfers to any evaluation that uses one. **Release.** The served node, the ten-model sweep rows together with the harness that generated them, the exposure decomposition, the production paired study and both of its judge passes, the negative-control completions with both their archived and their re-judged scores, the semantic re-score, the paraphrase stress set and the semantic audit (its sample, its cached judge calls, its summariser and the blind sample prepared for a human pass) are in the repositories footnoted below, under version control, with per-study bootstrap seeds named in the released scripts and a table-to-artefact manifest at `docs/research/gain-over-copy-paper/MANIFEST.md`. Every table and figure in this paper resolves to an artefact in that manifest. The release supports checking every reported number against saved rows; it does not support full scaffold regeneration, because the corpus-derived scaffold index is not released, and two embedding caches are excluded for size and rebuilt by their own released scripts against the embedding endpoint. The routing companion note (§16) states its own release gaps, which are not inherited here.

2 Related Work

Knowledge-graph and private-corpus grounding. Grounding lineage runs from RAG [29] through graph-structured retrieval [9, 19] and zero-shot knowledge-graph triple prompting [3] to roadmaps and surveys [42, 45]. GraphRAG addresses “private and/or previously unseen” collections [9], while enterprise variants highlight amortised curation [37]. Published work typically injects *instance* triples or opaque encodings,

including community summaries [9] and GNN soft-prompts over retrieved subgraphs [21]; we instead inject human-readable, schema-level content: class definitions, typed relations and taxonomy. None of these graph-grounding studies reports a standing input-exposure control beside its uplift figures, so their headline uplift is not re-centred on content already present in context, though where retrieval coverage is reported a reader can approximate the subtraction from their published quantities.

Baselines that reframe a result. Simple controls can overturn headline results: strong RAG baselines match elaborate pipelines under equal token budgets [27]; controlled comparisons reverse earlier “RAG wins” findings and add a confidence router [31]; and apparent argument comprehension can reflect spurious cues [41]. The confidence router of Li et al. [31] and selective retrieval [2, 23, 36] provide precedents for our injection gate. Those methods choose whether to retrieve; ours chooses whether to inject.

Input-only baselines and faithfulness metrics.

Our copy ceiling extends this tradition. Passage-only, question-only [25] and hypothesis-only baselines [20, 46] reveal shortcuts present in inputs; unlike these discrete task classifiers, the copy ceiling is continuous and per-item. Faithfulness can be measured separately from correctness [1, 11, 57]; metric-first evaluations formalise assumptions [38] and test shortcut robustness [14]. FActScore’s source precision and Adlakha’s token-overlap K-Precision resemble our exposure count, and ALCE rejects “copy the passage” as a shortcut where we use it as the reference point. Contamination audits [17, 48] assess input-visible score post hoc, whereas we define deliberate exposure as the reference a priori. The closest ancestry is the extractive-oracle and lead-*n* baselines of neural summarisation [39, 50], which likewise ask what a copy of the input already achieves.

Nearest neighbours, and how we differ. Closest in method is work that causally manipulates the gold answer to test its effect on retrieval-augmented gains [30]; that work is withdrawn, and we adopt only its paired-intervention *method*. Leakage-free benchmarks instead extract reasoning graphs from question-context pairs and apply type-constrained entity replacement with leakage filters [32], during corpus construction rather than as a per-item measurement. RAGChecker decomposes RAG quality into claim-level retrieval and generation scores through LLM extraction and grading [47]; our context utilisation is a deterministic, surface-matched version of its metric and we claim no originality for the decomposition. MIRAGE measures benchmark-level adaptability against oracle context across retriever-model pairings [44]. Judged arms use a cross-family judge to limit self-preference [43, 56].

Ground claimed, stated narrowly. Against the three closest instruments we claim narrower ground than a new construct: RAGChecker’s context utilisation already performs the exposure/delivery decomposition, via LLM claim extraction rather than a deterministic matcher [47]; the reader–context diagnostic already reports graded, judge-free answer presence in packed context, validated by intervention [4], and is the closest comparator overall; and Sufficient Context already shows that context-derivable-answer sufficiency is a stronger and better-validated criterion than raw answer-string presence [24]. Our contribution is an inexpensive, deterministic, judge-free diagnostic and a reporting convention: four-count accounting with a signed copy-baseline scalar, a copy-as-answer invariant, a seed-disjoint placebo, and a stratified audit that prices what the matcher gives up, together with the failure modes we document and the decisions they change. It is not priority over any one component, nor a general boundary between copying and reasoning.

3 Curated-Corpus Setting

This node is a served binary² operating over a corpus produced by a separate, CI-gated pipeline³ and accelerated by an in-process HNSW index⁴.

3.1 Ecosystem in brief

Knowledge enters through the *knowledgeGraph* repository: 8,138 Logseq markdown pages compiled losslessly into a pure-TBox OWL 2 ontology of 8,146 classes and no individuals, published as versioned generations. Content comes from human-directed authorship and, more recently, automated extraction through a preview-first import that refuses overwrites and emits review candidates rather than publishing machine output [18]. A CI gate combines consistency tests with a Whelk OWL 2 EL reasoner [5], rejecting contradictions before publication; each generation ships its reasoned closure of 282,492 triples.

Loom, the measured node, mirrors and serves a generation, combining lexical retrieval over class titles, confidence-gated injection of per-IRI markdown blocks, read-only SPARQL over the closure and delegation to the local model through an OpenAI-compatible façade, currently Qwen3.8-27B on two workstation GPUs. This is the subject of every measurement below.

Loom is one component of a larger stack, and two of the others matter here because the write-path results are only reproducible against them. Ingestion and the ledger writer of §13 run in the *VisionFlow* agent

²The served node <https://github.com/DreamLab-AI/loom> (code under the repository licence; corpus terms per the sibling *knowledgeGraph* repository).

³The canonical builder <https://github.com/DreamLab-AI/knowledgeGraph> (published at <https://narrativegoldmine.com>) runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

⁴RuVector: <https://github.com/ruvnet/ruvector>.

environment, where the podcast extraction of §12 executes; admission to the served corpus runs through the graph’s governed propose/approve path, the ontology-bridge component of the same stack. Figure 1 draws the read path around the measured node and §13 the write path: the instrument is corpus-general, but the write-path study is only checkable where the write path lives.

Behind these choices sit five deployment requirements: curation the organisation controls, human auditability at single-entity granularity, accelerators that may locate the served markdown unit but never stand in for it, LAN confinement with the model delegated by URL, and confidence-gated injection. Jointly they demand high recall where the corpus is authoritative, with attribution metadata whose precision is itself unmeasured (§15), and no regression on general questions the model already answers.

4 Ontology Scaffold

Our deployed serving node is a single static Rust binary built as a hexagonal workspace: a domain core defining the port traits, one adapter per backend, a scaffold-policy module and a thin HTTP façade. Only the domain core mints a served unit, so no adapter can return its own row, triple or vector as the payload, and golden fixtures pin retrieval, gating and scaffold serialisation byte-for-byte to a reference implementation.

Canonical unit and the corpus. The unit of service and review is a markdown-with-ontology block scoped to one IRI: curated research prose preceded by typed relations (`subClassOf`, `requires`, `enables`, `uses`, `relatedTo`, `contrastsWith`). The corpus is a single sha-addressable *generation*, here **8,146 concept classes** and **282,492 triples** in the Whelk-reasoned closure, mirrored atomically so the node serves a known, mixed-build-free version. Unlike approaches that reduce knowledge to opaque community summaries or GNN-encoded subgraphs [9, 21], this system keeps reviewable markdown as the durable asset and lets the accelerators only locate, rank or attest to it [16].

Together, the lexical matcher and confidence gate form the exclusive injection authority: a single policy scales the scaffold budget to the retrieval score and injects nothing below a defined threshold (§10.8). Generation is delegated through the model-swap seam to the configured backend URL, currently Qwen3.8-27B, and responses carry model identity, so §9 can vary the model while holding grounding constant, or vary the serving path while holding the model constant. The **2.02 ms** figure is the model-free lexical match alone; a complete grounded answer is dominated by generation, at $\approx 90\text{--}170\text{ s}$ median end-to-end across the Study 2 sets (§10.5).

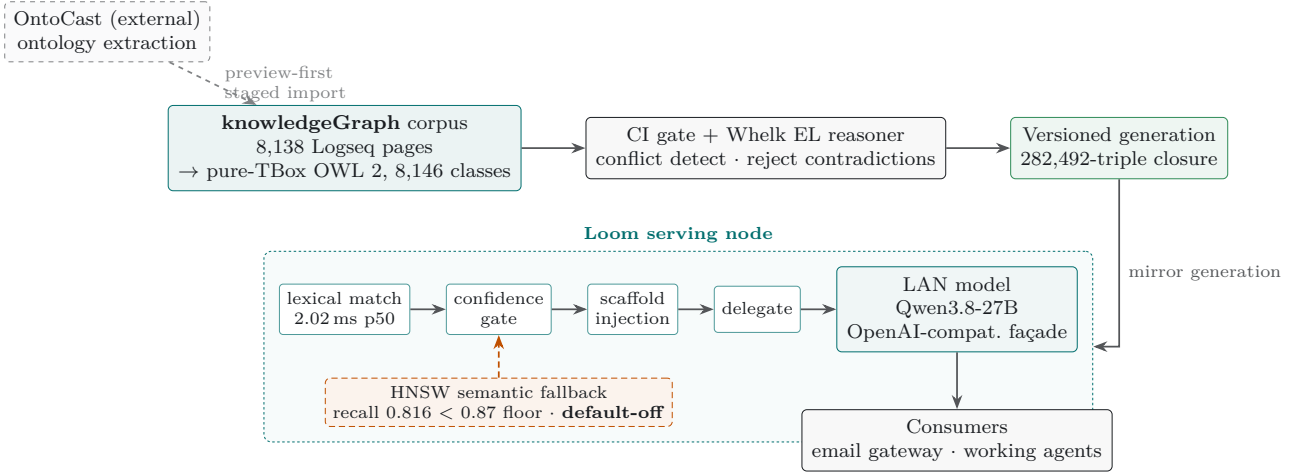


Figure 1: The data path around the measured node. Knowledge enters at the *knowledgeGraph* corpus (8,138 Logseq pages compiled losslessly into a pure-TBox OWL 2 ontology of 8,146 classes); output from OntoCast, an independent third-party extractor, enters only through a preview-first staged import that reports collisions and emits review candidates rather than publishing machine output directly. A CI gate and a Whelk OWL 2 EL reasoner classify each build and reject contradictions before publishing a versioned generation (a 282,492-triple reasoned closure). The *Loom* node mirrors a generation and serves it through a model-free pipeline (lexical match at 2.02 ms, confidence gate, scaffold injection) before delegating to whatever LAN model sits behind its OpenAI-compatible façade (Qwen3.8-27B here); consumers hold the façade URL and inherit grounding. The HNSW semantic fallback is drawn dashed in burnt orange because it ships default-off: its document-embedding recall (0.816) is below the 0.87 design floor, a red gate we report rather than hide (§3.1, §4).

Table 1: Request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate reaches the model, whether lexical or, when enabled, semantic.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 → seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units’ markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss → candidate seeds into the same gate	<i>default-off</i>
Delegate	Merge scaffold into the system message (last user turn is the retrieval query); call the configured backend URL	yes

5 The Copy Ceiling

We treat exposure accounting as the paper’s central contribution and the copy ceiling as its reportable scalar, and we therefore provide a formal definition of both, an explicit list of assumptions, and a dedicated robustness design, in line with the metric-first precedent [14, 38]. The order matters: the scalar is what a reader will quote, but the accounting is what survives a change of implementation.

5.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and answer a_i . Let $m(t, X) \in \{0, 1\}$ be the deterministic surface matcher, equal to one when the normalised title of gold item t , or at least 80% of its length- ≥ 4 words, appears in X . Define the exposed count and per-item **copy ceiling**,

which is a copy baseline or input-exposure reference and *not* an upper bound on admissible answers,

$$e_i = \sum_{t \in G_i} m(t, S_i), \quad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = (\sum_{t \in G_i} m(t, a_i)) / |G_i|$. A no-op extractor returning S_i verbatim scores exactly c_i . Every use of the name should be read as “the recall a copy already achieves”, never as a limit a good answer may not pass: a model can and sometimes does exceed it, and a perfect answer may sit exactly on it. Exposure is computed over the full visible input (the injected scaffold plus the question text); on the frozen sets the question text adds no exposure beyond the scaffold on any of the 510 questions, and for the raw arm (question text only) exposure is small but non-zero: 35 of 1,136 gold items appear in the bare question, a mean raw-arm ceiling of 0.037.

Scoring symmetry, stated and checked. One question template (T-COMMON, 45 of 510 questions) accepts *any* of several valid targets, and the released implementation computes the ceiling as the exposed *fraction* for these questions while the answer scorer credits any match. On the frozen sets this asymmetry is numerically inert, because every multi-target question’s alternatives are exposed all-or-nothing: the symmetric scorer reproduces every reported gain to four decimal places, and the executable invariant, that the concatenated visible input scored as an answer equals the reported per-item ceiling, holds on all 510 questions under both definitions and runs as a hard check in the released script. It does not, however, resolve the gold-set issue of §9: a question-level invariant cannot tell a required component from an interchangeable alternative.

Assumptions (when the instrument applies).

- (A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so gold is deliberately present in the injected text. The ceiling measures this exposure; it is not a leak.
- (A2) **Shared matcher.** c_i and r_i use the same matcher. The matcher is a surface proxy: it under-counts paraphrase, and it can also over-count (its exact-title path is a substring test, and its long-word threshold ignores word order, relation roles and negation, so naming a correct entity in a wrong or negated relation still scores). Absolute recall is therefore neither a strict lower nor a strict upper bound on semantic correctness. Applying one matcher to both sides means the two scores share their error sources, so some of that error may cancel in the paired difference $r_i - c_i$; how much is an empirical question, not a guaranteed order of robustness, because error rates can differ between structured scaffold text and generated prose. §10.2 measures the matcher’s error rates on both sides of the accounting, and §6 re-scores under a paraphrase-tolerant matcher.
- (A3) **Per-item, model-free ceiling.** c_i depends only on (q_i, S_i) and is identical across models given a fixed scaffold, enabling comparison across a model sweep.
- (A4) **An appropriate gold set and matching criterion.** What the ceiling requires is a gold target set that is the right answer to the question and a matcher whose criterion suits that set. Gold and context deriving from one source is not itself the requirement; it is the case in which the exposure confound is most acute, and therefore the case in which the reference is most needed. Where gold is independently authored (the out-of-domain arm), $c_i \approx 0$ and judged quality carries the axis.

A recall reference point, not an endorsement. Although the no-op extractor scores c_i , it is not an admissible answer: returning the entire scaffold lacks

the required precision, conciseness and relevance. The ceiling is therefore a recall reference for delivery, not a recommendation to reproduce the context.

5.2 Exposure accounting: the counts are primary

For each gold item $t \in G_i$ write $x_t = m(t, S_i)$ for *exposed* and $y_t = m(t, a_i)$ for *recovered*. Pooling over all gold items of a run gives four counts,

$$\begin{aligned} n_{11} &= \#\{x_t=1, y_t=1\}, & n_{10} &= \#\{x_t=1, y_t=0\}, \\ n_{01} &= \#\{x_t=0, y_t=1\}, & n_{00} &= \#\{x_t=0, y_t=0\}, \end{aligned}$$

with $N = n_{11} + n_{10} + n_{01} + n_{00}$. These four counts determine the micro-averaged exposure and recovery metrics listed in this subsection, and those alone; macro-averaged question scores, question-level uncertainty and judged outcomes need the per-question records, and different allocations of the same four counts across questions give different macro scores. The pooled ceiling is the exposed fraction $(n_{11} + n_{10})/N$, pooled recall is $(n_{11} + n_{01})/N$, and their difference, the pooled gain over copy, is

$$\bar{g} = \frac{n_{01} - n_{10}}{N},$$

the net of two cells. Context utilisation, $n_{11}/(n_{11} + n_{10})$, is the fraction of exposed gold that survives delivery; beyond-exposure recovery, $n_{01}/(n_{01} + n_{00})$, is the fraction of unexposed gold the model supplies itself. The headline figures in §10 average per-question ratios rather than pooling items, so they differ from these identities in the second decimal; the released decomposition script reports both.

Two properties make the counts the right object to report. They are diagnostic where the scalar is not: a gain of zero is consistent with $n_{01} = n_{10} = 0$ and with $n_{01} = n_{10} = 200$, a model dropping as much exposed gold as it invents, and only the counts tell those apart. And each x_t depends on the matcher and fixed text alone: no ranking, no threshold, no tuning. Where a task has no natural no-op extractor a baseline must instead be *constructed* from a ranker, which is a different instrument and not the quantity defined here (§16). The ceiling is the number to quote; the counts are the number to trust.

5.3 Gain over copy

We define the signed **gain over copy** as $g_i = r_i - c_i$, reporting the headline figure $\bar{g} = \bar{r} - \bar{c}$. It re-centres grounded recall against the exposure reference, and pooled over items it is $(n_{01} - n_{10})/N$ (§5.2).

- $g \approx 0$: the model recovers about as much as the context exposes. This is a *net*, not a statement of fidelity: it holds when $n_{01} = n_{10} = 0$, and equally when the model drops 200 exposed items and supplies 200 unexposed ones. Only the counts separate those two, which is why the counts and not the scalar carry the findings.

- $g < 0$, *lossy restatement*: the model drops more surfaced gold than it recovers unsurfaced gold. The decomposition of §10.3 measures the two cells directly, and here the deficit is almost entirely exposed-but-omitted gold (n_{10}), with n_{01} negligible. The magnitude places models on an exposure-normalised scale that raw uplift, which conflates exposure with delivery, does not provide.
- $g > 0$: the model recovers gold beyond what the copy baseline finds. The sign does not identify its source. Parametric memory, guessing, paraphrase that the matcher catches in the answer but not in the context, and matcher asymmetry all produce it, and none is separable from reasoning by this instrument.

A near-ceiling result shows the measured recall is achievable by delivery alone: the metric cannot credit recovery beyond exposure, and no reasoning is needed to explain the score. It does not measure how the model selected the correct names internally, so it constrains what the evaluation can attribute, not what the model did.

What zero gain does and does not say. The accounting is not a test for the presence of reasoning, and no result in this paper should be read as one. Consider a context stating that A requires B , that B has part C , and that D has part E , and a question asking for a part of something A requires. Choosing C requires following two relations, yet C is already a string in the context, so exposure is 1, a perfect answer scores 1, and gain over copy is exactly zero. Zero gain is therefore compatible with successful reasoning over exposed names, and negative gain with reasoning that sometimes fails; positive gain is compatible with memorisation, guessing or matcher asymmetry. The example generalises: wherever a curated corpus is served one entity-scoped block at a time, the names an answer needs are usually already on the page, and a recall metric over those names cannot see which route the model took to them. The defensible statement is the narrow one, *this recall score does not establish reasoning beyond answer-name exposure*, and establishing whether a model composes exposed structure needs a second instrument that varies the route while holding the answer name exposed. This paper does not run that instrument; it separates the two rather than conflating them, and §10.2 shows empirically where the exposure instrument stops, since roughly one credited item in sixty names the gold target without asserting the relation the question asked for. Figure 2 illustrates the two paths compared.

5.4 Negative controls: what would move the gain

A positive delivery result might reflect lexical overlap rather than knowledge transfer. Four arms hold the model and question fixed while perturbing only the injected block (defined here; results in §10.7):

true the block served by the node, injected into the bare model. It should reproduce the live loom arm, showing that the serving path adds nothing beyond the block.

shuffled the same block with body sentences shuffled. Whole-sentence shuffling preserves every within-sentence fact, so this arm tests sensitivity to sentence order only.

masked the same block with every seed-class title replaced by **Entity- k** . Answer-bearing *target* names remain, so this probes dependence on seed naming, not on answer-name matching.

irrelevant the well-formed, on-corpus scaffold of a *different* question, using a fixed derangement over engaged items. Seed-IRI disjointness is a construction check: it prevents donor and target sharing seed classes, but does not by itself guarantee that the donor block exposes none of the target’s gold, ancestors or synonyms (§10.7). Survival would show that any ontology-shaped text suffices.

Each arm answers a different question. *Shuffled* and *masked* ask whether sentence order and seed naming carry any of the effect. *Irrelevant* asks whether the specific retrieved content matters beyond the presence of a well-formed, on-corpus block: it is the content floor, and because it is a real scaffold for a real question it holds register and provenance fixed, though not token count. Together they test whether the observed lift is content-specific, which is the direct analogue of a shortcut-robustness check for a faithfulness metric [14]. Results are in §10.7.

5.5 Relation to prior instruments

Three recent instruments ask a related question, did the shown context deliver the answer, and §2 concedes precedent on each component. What remains to place is the position of this instrument among them. RAGChecker’s context utilisation performs the same decomposition through LLM claim extraction and entailment checking [47]; restricting it to a deterministic surface matcher buys zero judge cost and exact reproducibility, and costs semantic discrimination. The reader–context diagnostic is the nearest neighbour, scoring graded, judge-free answer presence in packed context [4]; the distinction is accounting and apparatus, not the underlying presence idea. *Sufficient Context* asks a strictly stronger question, whether *an* answer is derivable from the context at all, and its ablation shows that criterion beating a ground-truth-string-in-context baseline, 93% against 81% agreement [24]; our exposure ceiling is deliberately that weaker baseline. The idea of an input-side reference is older still: the extractive-oracle tradition in summarisation [39, 50, 53] makes a copy of the input the natural reference for any recall-style metric, and an information-bottleneck view compresses retrieved context against the answer for a related purpose [58]. The trade-off is explicit and is the reason §10.2 exists: removing semantic judgement removes semantic discrimination, so the ceiling is

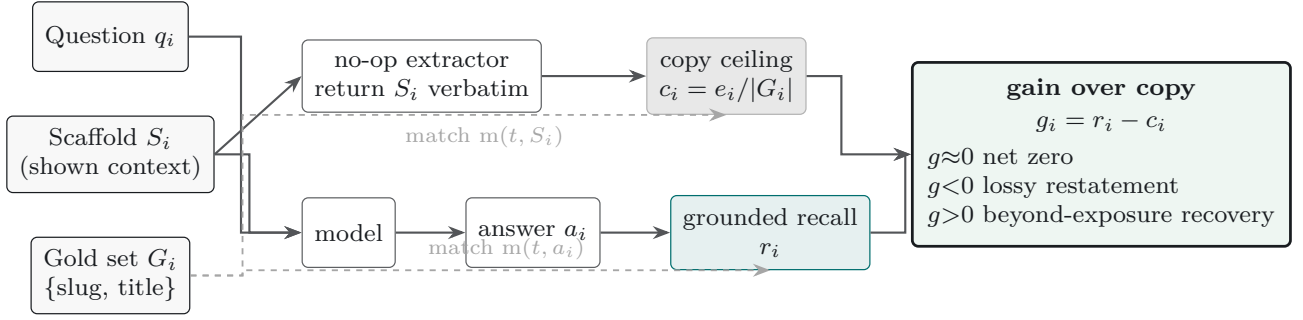


Figure 2: The copy ceiling and the signed gain over copy on one panel. The same shown context S_i feeds two extractors: a no-op extractor that returns S_i verbatim, whose recall against the gold set G_i is the per-item *copy ceiling* c_i ; and the model, whose answer a_i scores the grounded recall r_i . Both scores use the identical surface matcher $m(t, \cdot)$ with G_i as reference (dashed), so their difference, the *gain over copy* $g_i = r_i - c_i$, shares its error sources and some of that error may cancel; how much is an empirical question, since error rates can differ between scaffold text and generated prose (§10.2). The sign reads as a net, not as a verdict on fidelity: $g \approx 0$ means the model recovered about as much as the context exposed, which holds both when nothing was dropped or added and when omissions and recoveries cancel; $g < 0$ is net loss of surfaced gold; and $g > 0$ is recovery beyond what the exposure check finds, without identifying its source (§5).

read as a delivery-recall reference and the audit prices what that reading omits.

6 Semantic Re-Score: Is the Ceiling a Lexical Artefact?

Because the matcher is lexical, a sceptic can ask whether the negative gain is an artefact of strict matching: the answer paraphrases gold that the ceiling, computed on the same matcher, happens to catch. We re-score every saved sweep row with a paraphrase-tolerant embedding matcher, re-scoring the copy ceiling under the identical matcher so the comparison stays symmetric. The matcher embeds candidate spans (word 1–6-grams within delimiter-bounded chunks, plus each chunk and the whole text) with `bge-small-en-v1.5` and scores a gold item as recovered when the maximum cosine against its title meets a threshold; we sweep thresholds 0.80/0.85/0.90 and use the paper’s seeded 10,000-resample paired bootstrap. The copy contexts here are *reconstructed*, not shown byte-identical to the contexts of the original sweep: the released rows store each question’s exposure count rather than its context string. They are regenerated with the frozen scaffold engine and admitted only under an equivalence gate, which requires every question’s recomputed exposed-gold count to equal the count stored on its sweep row and the mean lexical ceiling to agree to four decimal places (0.9645). Two further quantities match the original run: the regenerated set yields 778 gold titles, the figure that run recorded, and a 126-item sanity sample of lexical hits agrees with semantic scoring at 100% at every threshold. That gate is an equivalence check on exposure and on lexical scoring. Two texts with equal lexical exposure may still be worded differently, and embedding-match behaviour could differ between them, so this is a re-score on reconstructed contexts rather than on the original bytes.

Level moves with the threshold; sign does not. At 0.80 and 0.85 semantic scoring raises both recall and

ceiling, the ceiling reaching 0.980 and 0.968 against 0.9645 lexical; at the strictest threshold, 0.90, both fall below their lexical values (0.937 ceiling). In all **thirty** model $\times$ threshold combinations the gain over copy stays negative with the 95% CI excluding zero. Per-model magnitudes move slightly, the largest shift being Qwen-2.5-72B from -0.050 lexical to -0.034 at 0.80, but no model’s interval crosses zero. Pooled beyond-exposure recovery stays at 3 of 11,360 at both 0.80 and 0.85, where the pooled decomposition is $n_{11} = 10,001$, $n_{10} = 669$, $n_{01} = 3$, $n_{00} = 687$; only at 0.90, where the matcher has become strict enough to lose exposure as well as recall, does it reach 20, and no single model exceeds 4 at any threshold. The headline is therefore not an artefact of lexical matching.

Unlike the lexical instrument, this check depends on an embedding model and its span extraction; it is a robustness probe on the sign of the gain, not a replacement scorer, and it speaks to lexical strictness rather than to relational correctness, which is the subject of §10.2.

7 Making the Copy Ceiling Runnable

This control is cheap to adopt because it needs no model. To compute it on any corpus:

- Inputs.** A list of questions, each with a gold target set (the {slug, title} pairs the source asserts as its answer) and the exact context string S_i that would be shown to the model. The ceiling needs a gold target set appropriate to the question and a matching criterion suited to it (assumption A4); shared provenance between gold and context is not required, though it is the case where the exposure confound bites hardest.
- Matcher.** A deterministic surface matcher $m(t, X)$: one when the normalised title of gold item t , or $\geq 80\%$ of its length ≥ 4 words, ap-

pears in X . The reference implementation and the scaffold engine used here are released in the harness (`tools/paper/decompose_exposure.py` and `ontology_scaffold_v1.py`); a paraphrase-tolerant embedding variant is a drop-in (§6).

- 3. Compute, no model call.** Per question, the copy ceiling is $c_i = (\sum_{t \in G_i} m(t, S_i)) / |G_i|$: the fraction of gold already exposed in the context. The headline is the mean over questions. Grounded recall r_i reuses the same matcher on the model’s answer; gain over copy is $r_i - c_i$.

Reading the number. A ceiling near 1 (0.964 here) says the context already contains the answer strings, so retrieval supplies most of the coverage. The exposure reference then belongs *beside* the quality and system baselines a generative component is judged on, not in place of them: positive gain over copy is not a prerequisite for useful generation, and a component that restates a curated source faithfully is doing its job at a gain of zero. A ceiling near 0.5 or below says the answer strings are largely missing, which may mean the task genuinely requires composition, or equally that retrieval failed, that the question was phrased out of vocabulary (§8), that gold is incomplete or that the matcher missed. In both directions the number is a prioritisation heuristic that tells you where to look first, not a classifier of what the model is doing, and it never says that the required relation is present. A worked question, its gold and a sample scaffold block are in Appendix B.

8 Vocabulary Mismatch: the Retrieval Boundary

Our serving node retrieves lexically, and its red-gated semantic fallback exists for the case where a question is phrased outside the graph’s title vocabulary. We test that case with a paraphrase stress set. Each of the 510 sweep questions is rewritten by `gpt-4.1` at temperature 0 under seed 42 to preserve meaning while excluding the seed class’s title tokens, and each candidate passes a three-part gate before entering the set: no gold title appears verbatim; no content token of the seed class’s title appears by exact or prefix match; and the asked relation is preserved, adjudicated by the same model. Candidates are retried up to four times. **506 of 510** questions clear the gate at a mean of 1.112 attempts, four remain unresolvable, and the final set contains zero title-token leaks by construction. The 61 rejected attempts divide into 33 that named the gold verbatim, 23 that failed to preserve the relation and 5 that leaked a seed-title token.

Here the instrument is the copy ceiling itself, computed per arm with no generation: does the retrieved block expose the gold? Two arms are compared. The *lexical* arm queries the live production façade. The *semantic* arm takes top-5 cosine seeds over all 8,146 class embeddings and feeds them into the same frozen scaffold engine at the same budget. The façade’s own

semantic path is not reachable: `/health` reports it unready because the deployed index artefact declares no embedding model and fails the contract check, so the semantic arm is necessarily a harness-side reconstruction over the same corpus and the same embedding model.

Exposure collapses. Mean ceiling falls from **0.964** on the original questions to **0.328** on the paraphrases, with **64.4%** of questions (326 of 506) below 0.5. The live façade and the frozen engine agree to four decimal places on the original questions, which confirms that the deployed node retrieves what the frozen harness does.

Silent failure. Because partial word overlaps almost always engage some seed, the designed absence-keyed fallback fires on only **2 of 506** questions (0.40%), while 326 have a ceiling below 0.5. A gate keyed to finding nothing cannot detect finding the wrong thing, so the fallback as designed is inert against the failure mode that occurs.

A real signal, wasted by the gate. The semantic arm lifts the mean ceiling to 0.596, a recovery of +0.269 [+0.227, +0.310] over the whole set and **+0.446** [+0.398, +0.493] on precisely the 326 questions where lexical retrieval fails. The information needed to rescue those questions is therefore present in the corpus and reachable by a path the shipped gate never invokes.

Prescription, and its bounds. The specific, supported prescription is that a hybrid gate must be keyed to retrieval *confidence* rather than to retrieval *presence*: presence is satisfied on 99.6% of paraphrases while exposure is lost on 64.4% of them. Whether semantic seeds should then *replace* lexical ones or be appended to them under a fixed token budget is not measured here, and three considerations argue against replacing them wholesale. On entity and rare-token queries dense retrieval is the fragile arm and lexical matching holds up [49], so the safe form gates on query type or on dense-versus-lexical rank disagreement and fuses the rankings [6]. Score-distribution predictors are weakest precisely on semantic queries and cannot see a confident-but-wrong lexical seed [12], so the informative key is disagreement between the two retrievers rather than a single-ranking confidence score. And the collapse reported here is a retrieval-*exposure* measurement, whether the block reaches the model at all, not the model’s answer robustness: the paraphrase-robustness literature reports single-digit accuracy drops [35] because it measures a different construct downstream of successful retrieval, and the two figures are not comparable. Within the paper’s framing the ceiling again does the work: it shows where exposure is lost, measures what a semantic path could recover, and prices the design that would recover it.

9 Method

Design. Both studies use within-model pairs: the *same* model answers the *same* question twice, cancelling between-question difficulty variance. The sweep varies the *model* with grounding fixed; the production study varies the *serving path* with the model fixed.

Study 1: the ten-model sweep. The knowledge graph supplies templated questions for every sufficiently-defined class: **T-REL** (requirements, uses, dependencies, enablers, and parts), **T-TAX** (immediate and ancestral parents), and **T-COMMON** (shared ancestors of a pair). Gold answers are the {slug, title} targets asserted by the graph. Seed 42 yields **510 questions** across 15 domains: 285 T-REL, 180 T-TAX, and 45 T-COMMON. Each is posed *raw* and *scaffold* (with a budget-clamped 1,500-token ontology extract prepended) to ten models from eight developer families (Google, Anthropic, OpenAI, Zhipu, DeepSeek, Meta, Alibaba, Mistral), accessed through their providers’ APIs or a router, under one fixed configuration (`temperature=0`, `max_tokens=2048`; `reasoning_effort=low` only where thinking cannot be disabled, preventing mandatory passes from truncating answers). The lexical matcher from §5 counts a gold item when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. The matcher is a surface proxy that can both over-count and under-count, so absolute recall is not a bound in either direction (A2); the signal is the explicitly lexical *paired* delta for the same question, scorer, and model. Truncation is detected per row: GLM-4.6 truncated on 174 rows, of which 5 were retried and the rest scored as received, making that row lower-confidence, and Gemini 2.5 Flash-Lite on 16.

What the scorer does and does not validate. Two properties of the instrument bound every number the sweep reports. First, the matcher tests for the presence of a gold target’s name. It does not test that the answer asserts the right relation, that subject and object are the right way round, that the answer is not simultaneously denying the relationship, or that it is not an unusably broad list; an answer can therefore score fully while being relationally wrong. Using the same matcher on context and on answer does not guarantee that such errors cancel, because structured scaffold text and generated prose differ in length and syntax. §10.2 measures this gap directly on a stratified sample of the sweep and reports both the rate and the mechanisms. Second, the gold sets are not uniformly conjunctive: T-COMMON (45 of 510 questions, of which 17 carry two or more acceptable targets) accepts *any one* of several valid targets, while the item-level decomposition counts each target separately. A model can therefore give a fully acceptable answer and still accumulate exposed-but-omitted items, so item-level omission counts include some compliant omissions; §10.3 recounts them and §10.2 checks that recount against judged assertions. Separating required

components, interchangeable acceptable answers and optional additions in the gold sets is outstanding work.

Study 2: the production-node paired study. We hold the model constant at Qwen3.8-27B on the reference GPU host, isolating the serving path. The *loom* arm calls the production Rust node’s `/v1/chat/completions` endpoint, which retrieves, confidence-gates and injects the scaffold, returning telemetry (engagement, injected tokens, fusion path) in the response *loom* block. The *raw* arm calls the same backend without a scaffold. Both use `temperature=0` and `max_tokens=1536`, below which the reasoning backend truncates output to empty. Three pre-generation frozen sets contain **117 questions**: 24 *arcane* (deep in-domain), 33 *thin* (sparsely covered), and 60 *general* (out-of-domain). The general set is also graded by the out-of-domain judged arm below, keeping the judged arms comparable. The negative-control arms from §5.4 use the same node; their scaffolds come from the LLM-free `/loom/scaffold` endpoint, so each control presents the bare model, via a system message, with exactly the block bytes served to the *loom* arm.

Out-of-domain judged arm. A five-model judged arm poses the 60-question *general* set (adjacent, in-domain-general and off-domain subsets) to *harnessed* and *bare* configurations. Each pair is graded 0–5 against frontier-authored key points, providing an independent out-of-domain instrument (§10.8).

Cross-family semantic judge. To cover paraphrase across Study 2’s in-domain and out-of-domain questions, we use a reference-guided **0–5 rubric** judge rather than lexical scoring. The judge is `openai/gpt-4.1` via OpenRouter with `temperature=0` and `max_tokens=200`, applying the same fixed system-message rubric in every arm. It is from a different family than the Qwen candidate, limiting self-preference [43, 56]; it is blind to each answer’s arm and study origin, and grading can resume per (set, id, arm). References are graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. Per set and pooled, the primary metric is the paired mean difference with a seeded **10,000-resample bootstrap** 95% percentile interval [10, 26]. Alongside it we report a two-sided **exact conditional signed-rank** test, which targets a rank summary rather than the mean and is therefore complementary: zero differences are dropped (the reduced-sample procedure, not Pratt’s), tied absolute differences receive average ranks, and the *p*-value comes from exact enumeration of the sign-randomisation distribution conditional on the observed ranks, which matters because one set has only two non-zero pairs. The paired effect size is the **matched-pairs rank-biserial correlation** $r = (T^+ - T^-)/(T^+ + T^-)$ over non-tied pairs.

Set-level p -values receive a **Holm–Bonferroni** correction [8]. For out-of-domain non-regression we use a conservative *interval-inclusion* criterion in the spirit of equivalence testing [28]: non-regression is declared only when the 95% bootstrap percentile interval lies wholly inside a pre-specified ± 0.25 margin, which is stricter than the 90% interval conventional TOST would use. Because in-domain graph-derived questions cluster by class and domain [8], we report a domain-clustered block bootstrap alongside the naive interval and an intention-to-treat delta retaining questions where the gate did not engage [22].

10 Results

Budget exhaustion and the complete-case sample (Study 2). At `max_tokens=1536`, 42 of 234 loom/raw completions, spanning 28 question pairs, exhausted the budget during reasoning and produced no output. Each affected pair was re-executed at 4096 in *both* arms; three thin questions still came back empty in one arm (loom once, raw twice, so realised missingness was not symmetric) and were excluded pairwise, giving a complete-case $n = 114$. The pooled contrast is therefore a mixed-budget estimand: 25 re-run pairs at 4096 and 89 at 1536. The control arms of §10.7 remained at 1536 and lost 36–50% of their completions the same way, which is accounted for there.

10.1 In-domain: recall against exposure

For Gemini 3.7 Flash, raw recall is 0.375 and scaffold recall 0.942, a paired delta of **+0.567** with naive and domain-clustered 95% CIs of $[+0.532, +0.603]$ and $[+0.524, +0.611]$, respectively, across $n = 510$ items; there were no errors or truncated responses. The low raw score is consistent with a domain only partially covered by pretraining (the corpus is public, so some exposure is expected; §1). The copy ceiling is **0.964**, placing recall -0.022 below the question-averaged exposed fraction. (That macro ceiling is not the item-pooled exposed fraction, 0.933; the two are reported separately and are not interchangeable, §17.) The model therefore returns most curated answers and omits some. Because recall does not measure additions beyond the gold, fabrication and attribution precision remain unmeasured (§15); §10.2 measures whether a credited name is asserted in the relation the question asks for.

10.2 Semantic audit of the matcher

Our matcher tests for the presence of a gold target’s name. It does not test that the answer asserts the right relation, that subject and object are the right way round, or that the answer is not denying the relationship. This section measures how far that gap moves the numbers, on the released sweep itself.

Frame and sample. The unit is a (question, model, gold item) triple in the scaffold arm, and the frame is every such unit in the ten-model sweep: 510 questions, ten models, 11,360 units. Regenerating the exposure classification from the frozen question set and the scaffold index reproduces the released decomposition exactly ($n_{11} = 9,865$, $n_{10} = 735$, $n_{01} = 3$, $n_{00} = 757$), which is the gate the sample is drawn behind. Sampling is stratified by matcher category, then within each category over (template $\times$ gold type) cells proportionally to cell size, then over models within each cell, then simple random sampling without replacement, under seed 42. The result is **423 units**: 240 of the 9,865 credited-and-exposed, 120 of the 735 exposed-and-omitted, 60 of the 757 unexposed-and-uncredited, and a census of all 3 unexposed-and-credited. Exposure is model-independent, so the unexposed strata are 76 distinct items repeated across ten models and are not independent observations.

Adjudicator. Each unit is put to `openai/gpt-4.1` at temperature 0 through OpenRouter, with the question, the gold edge, the full gold set, whether the question requires all or any one of it, and the answer. The rubric asks for one of six labels: *correct_relation*, *name_only*, *wrong_relation*, *reversed*, *negated*, *absent*, with a fixed precedence order, plus whether the item is required or an accepted alternative, a confidence, and a verbatim quote. All 423 units returned a parseable verdict, two after a single formatting re-ask; every call is cached by a `sha256` of judge model, rubric and prompt, so re-running returns identical verdicts. The total adjudication cost was \$0.93. **The adjudicator is a model, not a human.** It is cross-family to nine of the ten swept models and the same family as the tenth, `gpt-4.1-mini`, whose row therefore carries a self-preference risk the others do not; the per-model breakdown is released so a reader can check it, and the blind 120-unit sample and annotation guide for a human pass are released unannotated.

Quote gate. Every non-*absent* verdict must cite a verbatim span of the answer, and the audit checks that span under the matcher’s own normalisation. The policy is one policy for every failed citation, and it never consults the matcher’s own category, so credited and uncredited strata are treated alike. A verdict whose cited span cannot be found in the answer is put back to the judge once, with the rubric and the item unchanged, asking for a span that occurs verbatim; a verdict that still fails to supply one is marked *unresolved* rather than silently kept or silently demoted. Of 281 verdicts required to cite, 252 verify at the first pass, and the 29 that do not are adjudicated under that one policy: 11 are repaired by a re-asked span that verifies, and on the other 18 the judge conceded that no supporting span exists. Repairs and concessions fall along the strata: all 11 repairs are in the credited stratum, 6 verifying character for character and 5 under the matcher’s own normalisation; all 18 concessions

are in the uncredited strata, 14 in n_{10} and 4 in n_{00} . A re-asked span is accepted when it matches the answer character for character, or under the matcher’s own whitespace and punctuation normalisation, which keeps the gate no stricter than the policy it replaces; both flags are recorded per unit in the released log. Under a strictly verbatim rule the five normalisation-only repairs would be unresolved instead, all five in the credited stratum and all carrying a *correct_relation* verdict, and credit precision would read 0.950 (228/240) against the matcher, 0.971 (233/240) for it and 0.970 (228/235) with them excluded. Every rate below is read three ways, which bounds what the unresolved verdicts can do to it: taking each unresolved verdict at its word, counting it as *absent*, and dropping it from its stratum’s denominator. Under the acceptance rule as run, credited-item rates are invariant across all three, precision of credit holding at 0.971; the omission rates move, and are quoted with that spread where they appear.

Precision of credit. Table 2 gives the confusion (upper panel). Of the 240 credited items, 233 are judged to assert the gold edge: precision of credit **0.971** [0.941, 0.986]. Four (0.017 [0.006, 0.042]) name the target without asserting the asked-for relation, one (0.004) places it in the wrong relation, and two (0.008) the judge cannot find at all. The credit the matcher issues is therefore reliable at roughly nineteen items in twenty asserting what the question asked, with the residual split between bare naming and outright relational error about four to one. Two different questions arise on the unexposed side, and the audit answers them with different evidence. The first is whether the matcher hides correct answers it never credited: on 60 unexposed-and-uncredited items the judge finds no hidden correct relational answer (0 of 60, 0.000 [0.000, 0.060]), which rules out an undetected-recovery rate above about 6% on this sample. Those units say nothing about false credit, because the matcher issued no credit on them; they are uncredited by construction. False credit is answered only by the second question, and only by the n_{01} census: all three unexposed-but-credited items in the entire frame fail relational adjudication, one naming the target without asserting the edge and two the judge cannot locate at all. Frequency and precision are separate facts here, and both are needed. Credit beyond exposure is rare, three units in 11,360; and on this census, which is every such unit there is, its precision is zero. The three-item beyond-exposure recovery of §10.3 therefore survives the audit as noise rather than signal, though three units support no rate estimate.

Correcting the exposed-item rate. Reweighting each matcher stratum by the judged probability that the answer asserts the gold edge moves the exposed-item rate from the matcher’s context-utilisation figure of 0.931 to an estimated **0.905** [0.884, 0.923], a difference of 2.5 percentage points. This is an estimated

correction under this adjudicator and this protocol, and not a known overstatement. Three qualifications belong beside the number. The adjudicator is a model and the demotion rule is a quote policy, so both the size and the sign of the correction inherit the judge’s own errors and that policy’s choices. Both strata are sampled rather than censused, and their observations repeat questions and gold targets across ten models, so they are not independent draws. And the denominator is lexically exposed items while the judged quantity is whether the answer asserts the gold edge, which is a property of answers, not a measure of whether the shown context was sufficient to produce them. The estimator is the frame-weighted combination of the two judged stratum rates, $R = (9,865 p_{11} + 735 p_{10})/10,600$ with $p_{11} = 233/240$ and $p_{10} = 3/120$, and its interval comes from a stratified cluster bootstrap that resamples *questions* within each stratum, 184 of them in the credited stratum and 59 in the omitted one, re-computing both rates on every draw and recombining them through the same weights, at 10,000 draws and seed 42. Resampling questions rather than units is what carries the dependence between gold items of one question; clustering on model instead of question gives [0.886, 0.924], so neither crossed factor dominates. Propagating the two Wilson intervals naively gives [0.876, 0.922], which is reported for comparison only and is not a valid interval for this design: it adds bounds instead of resampling the joint distribution, and it ignores the clustering altogether. Across the three readings of an unresolved verdict R moves only from 0.9052 to 0.9054, so the correction does not turn on how those units are counted. The gap is small partly because the two error directions offset: credited items are occasionally not asserted, and omitted items are occasionally asserted after all. That offset is a fact about this corpus and this matcher, not a general property.

Omissions. This is the stratum the unresolved verdicts live in, so its rates are the ones that move under the three readings. Of the 120 exposed-but-omitted items, true omissions, the target absent and nothing substituted for it, run from 0.708 to 0.825 across those readings (0.802 when unresolved units are excluded), and the share where the judge finds the item named at all runs the other way, from 0.225 down to 0.108 (0.123 excluded). Three (0.025 in every reading) are outright matcher false negatives where the judge finds the edge asserted; five (0.042) are answers that explicitly *deny* the edge, which the matcher and a true omission cannot distinguish; and between five and two (0.042 to 0.017; 0.019 excluded) supply an accepted alternative instead. That last figure recounts the compliant-omission correction of §10.3 from the other side. The deterministic any-collapse rule flags 52 of 735 omissions (0.071) as compliant; the judge, requiring the substitute to be genuinely asserted rather than merely matched, is stricter. The 52/735 figure should therefore be read as the upper end of that correction, not its point estimate.

Why the matcher errs. Three mechanisms are visible, and the first two are countable over the whole frame without any judge (Table 2, lower panel). *Subject-substring credit*: on 208 credited units the gold title is a substring of the question’s own subject name, so merely restating the question earns credit. This is the weakest credit path in the audit: of the seven such units sampled, four are judged *correct_relation* against 0.971 overall. *Run-together gold titles*: seven gold titles in the frozen set write their words without spaces (DeFi, EdDSA, GraphQL, MultiAccessEdgeComputing, OpenXR, SustainabilityReporting, WebRTC). They are not unmatchable. Of the 70 frame units carrying one, the matcher credits 58 and omits 12 (Table 2, lower panel), and all 58 are credited by a verbatim substring hit: the model reproduced the run-together form the scaffold put in front of it, so the normalisation’s refusal to split the title never bit. The blind spot is to the spaced expansion, not to the title, and it bites on 10 of the 12 omissions, where the answer does contain the spaced form and the matcher cannot see it (“Sustainability Reporting” for SustainabilityReporting); the other two do not mention the item under any spelling, and there the matcher is right. The defect therefore costs 10 units of 11,360, from two titles: SustainabilityReporting on one question for nine of the ten models, and MultiAccessEdgeComputing once. The remaining five are conventional unspaced names in their fields, which is why models reproduce them exactly and the matcher never misses them. All seven sampled units carrying the mechanism are judged *correct_relation*. *Taxonomic attachment*: credited-item precision is lowest on T-TAX, 0.938 [0.832, 0.979] against T-REL’s 0.978 [0.944, 0.991], and the wrong-relation case is a model naming a correct ancestor while asserting the wrong attachment point in the hierarchy. A matcher that tests for a name cannot see the difference between a parent and a grandparent.

What this licenses. The proxy is useful for measuring how much exposed gold survives delivery, at an estimated 2.5-point overstatement under this adjudicator, with credit on exposed items precise at 0.971; on unexposed items it issues credit three times in 11,360, and every one of those three fails adjudication. It fails not at random but on subject-name substrings, on the spaced expansion of run-together gold titles, and on taxonomic attachment. It does not license reading recall as relational correctness anywhere in this paper, and it does not replace a human pass, which is prepared and not performed.

10.3 Ten models: gain over copy

Table 3 applies the paired design to ten models from eight developer families under one fixed configuration. Raw→scaffold uplift ranges from +0.57 to +0.78, while scaffold recall spans 0.897–0.942 against the model-independent ceiling of **0.964**. Hence raw uplift does not distinguish delivery from exposure.

Table 2: *Top*: semantic audit confusion, matcher category against model-judge verdict, as judged (before the quote gate); *Name* is *name_only*, the target named without the asked-for relation, *Wrong* pools *wrong_relation*, *reversed* and *negated*, and *n₀₁* is a census of all three units while the other rows are stratified samples. *Bottom*: deterministic mechanism census over all 11,360 frame units, needing no judge; each row counts units carrying a structural property of the matcher or the question set, and the sampled verdicts then say how often each one changes the answer. Run-together gold titles are matched when an answer reproduces them verbatim and missed when it expands them into spaced words, which is why they appear in both the credited and the omitted columns; the blind spot is a property of the frozen question set rather than of any model.

Matcher category	Correct	Name	Wrong	Absent	<i>n</i>
<i>n₁₁</i> exp., credited	233	4	1	2	240
<i>n₁₀</i> exp., omitted	3	24	7	86	120
<i>n₀₁</i> unexp., credited	0	1	0	2	3
<i>n₀₀</i> unexp., omitted	0	6	2	52	60

Mechanism	<i>n₁₁</i>	<i>n₁₀</i>	<i>n₀₁</i>	<i>n₀₀</i>
Gold title inside the subject name	208	2	0	0
Credit via the ≥ 0.8 overlap path	28	0	3	0
Gold title runs words together	58	12	0	0
Gold title is a single word	1862	128	0	80
<i>Denominator</i>	9865	735	3	757

Signed **gain over copy** is negative throughout (Figure 3), from -0.067 for Gemini 2.5 Flash-Lite to -0.022 for Gemini 3.7 Flash. Because $g_m = R_m - 0.964$, ranking by gain equals ranking by scaffold recall; its contribution is an interpretable scale, not a new ordering. Recent models sit nearest the ceiling, while Gemini 2.5 Flash-Lite, GPT-4.1-mini and Mistral-Small-24B fall further below, an ordering specific to this task. Two models had transport-level truncations under the fixed 2048-token budget: GLM-4.6 had 174 truncated and 5 retried, with the rest scored as received and its row treated as lower-confidence; Gemini 2.5 Flash-Lite had 16 truncated. The 174 truncated GLM-4.6 rows are a substantial qualification on any reading of this table as a model ranking: the configuration is a valid fixed operating point, not a clean comparison of capability.

What adds information: the exposure/recovery decomposition. A negative g alone cannot distinguish omission of exposed gold from failure to recover unexposed gold. We therefore classify each item by scaffold exposure and answer recovery. Across ten models and 11,360 gold-item observations (the same 1,136 target instances scored under each of ten models, not 11,360 independent facts), $n_{11} = 9,865$ were exposed and recovered, $n_{10} = 735$ exposed and omitted, $n_{01} = 3$ unexposed and recovered, and $n_{00} = 757$ unexposed and omitted (Table 4). Thus only **3 of 11,360** items (0.026%) were recovered without exposure, with no model exceeding $n_{01} = 1$. With $n_{01} \approx 0$, $g \approx -n_{10}/|G|$: the deficit chiefly measures imperfect copying, including lexical under-counting

of paraphrase, rather than recovery beyond exposure. Context utilisation, $n_{11}/(n_{11} + n_{10})$, ranges from 0.900 to 0.955. Roughly one exposed item in fourteen is dropped ($735/10,600 = 0.069$, one in 14.4). That raw count includes omissions that are correct compliance with any-one-of questions: 166 of the 735 belong to any-type questions, and 52 of those are compliant, the model having recovered a different accepted alternative that the paper’s own scorer credits in full. Excluding them gives $683/10,600 = 0.064$, one exposed item in 15.5, and the two models with the weakest utilisation account for 24 of the 52. The adjusted figure is a *target-name omission rate*, not a bound on semantic error: the matcher can also credit relationally wrong answers and miss other error types, and the audit of §10.2 estimates the semantic rate directly. The headline recall figures are unaffected, since they already apply the any-collapse at question level; only the item-level drop rate moves. Utilisation is micro-averaged over exposed items whereas recall averages per-question ratios with varying c_i ; accordingly the rankings differ once, llama-3.3-70b having utilisation 0.934 against deepseek-chat’s 0.931 but recall 0.916 against 0.924.

10.4 Grounding suppresses parametric recall

This result needs no ceiling to state: it is a bare-versus-grounded comparison on exactly the $n_{01} + n_{00}$ items the scaffold does not expose, and it has the most direct operational consequence of anything in the paper. Bare, those items are not beyond the models: pooled over the ten models and the 760 unexposed gold items (76 per model), the raw arm recovers **12.1%** of them from parameters alone (92 of 760). Under injection the *same* items are recovered at **0.4%** (3 of 760), and seven of the ten models drop to *exactly zero*. Recovery without injected context therefore exceeds recovery with it on these items: the observed behaviour is suppression. Displacement of parametric knowledge by injected context, a documented tendency [34, 51, 55], is the natural explanation, though absence of an item from the scaffold is omission rather than contradiction, and this design observes the effect rather than isolating its mechanism.

Exactly what the tested items are. The items are gold targets *from this corpus* which the *particular retrieved scaffold* for their own question happened not to expose, assessed under *this* authority instruction, *this* confidence-gated retrieval policy and *this* lexical matcher. “Absent from the retrieved scaffold” is not “outside the corpus”, still less “arbitrary out-of-domain knowledge”, so nothing here licenses a claim about how a grounded model handles knowledge the corpus does not contain. Obeying an authoritative supplied source can also be the desired behaviour, making this a cost to price rather than a defect to fix. Three caveats bound the reading: the shared lexical matcher makes both rates surface proxies for paraphrased recovery; a bare-arm hit does not prove purely parametric provenance,

since lexical cues in the question can contribute; and the effect is a property of this injection policy rather than of grounding in the abstract.

Are the bare-arm hits real? The 92 raw-arm recoveries are lexical matches, so the gap the audit of §10.2 opens on the scaffold arm opens here too: a bare answer can name a gold target without asserting the relation the question asked for. All 92 were therefore put to the same relational adjudicator, under the same rubric and the same quote policy. The judge calls 71 of them *correct_relation* (0.772 [0.676, 0.846]), 15 a bare naming and 6 absent, with none placing the target in a wrong, reversed or negated relation; three citations needed adjudication and one of those stayed unresolved, which moves the rate no further than 0.780. Counted as relationally validated lexical recoveries, the comparison is 71 of 760 bare against 0 of 760 grounded. Adjudication costs the bare arm 21 of its 92 credits and the grounded arm all three of its three, so the direction of the contrast survives on the validated credits; per model the bare arm’s validated count runs from 1 of 1 to 13 of 14. Because lexical negatives were not exhaustively adjudicated in either arm (60 of the 757 grounded negatives were sampled, none of the 668 bare negatives), these counts do not establish full-frame relational recovery rates, and no interval is attached to them. The contrast stated above is between two lexical scores produced by the same matcher on both arms and is unaffected by this audit; what the audit changes is how much asserted parametric knowledge the bare figure stands for.

Scope of the observation. The figures are computed with the paper’s own byte-identical matcher over the released sweep rows (per-model raw recovery 1–14 of 76). This is the more informative of the two observations: supplying the context is followed by loss of recall the bare model demonstrably has, a stronger statement than grounded recall merely failing to exceed a verbatim copy. The design observes the behaviour without separating its mechanisms, and the wrapper ablation that would separate them is named in the conclusion’s future work.

10.5 Production-node paired study

Study 2 holds Qwen3.8-27B and decoding constant on the reference host while varying only whether requests pass through the shipped Loom node (*loom*) or reach the backend directly (*raw*). The node includes live retrieval, confidence gating and injection authority; a cross-family GPT judge scores answers 0–5 as described in §9. Table 5 reports paired results.

Across $n = 114$, loom raises judged quality by **+0.27** points ([+0.11, +0.45], exact signed-rank $p = 0.0023$, matched-pairs rank-biserial $r = +0.589$). The bootstrap interval targets the mean difference while the exact p targets a rank summary of the paired differences, so the significant p is not itself a test of the mean effect; they are complementary readings of the

Table 3: Ten models, one fixed configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall – copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

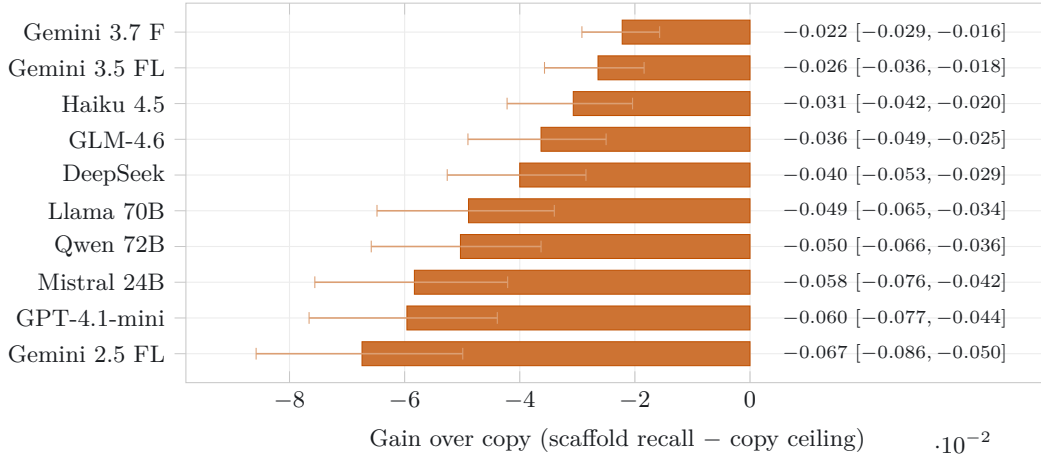


Figure 3: Signed gain over copy for all ten models, negative throughout, with per-model seeded 10,000-resample bootstrap 95% intervals (whiskers); the value column at right gives each point and interval, rounded independently from the full-precision artefact (points, whiskers and labels are all generated from it). No model’s interval crosses zero, and none exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) puts the models on an exposure-normalised scale that raw uplift (+0.57 to +0.78) does not provide. Models nearest zero (top) drop the least exposed gold. A value near zero is a net and not a statement of fidelity: it is equally consistent with no omissions and no recoveries, and with omissions and recoveries that cancel. Only the counts of Table 4 separate those, and the scale says nothing about the precision or relational correctness of what each model returns.

same 114 pairs. There were 24 wins, 8 losses and 82 ties, so the pooled effect rests on the 32 non-tied pairs, the other 82 scoring identically in both arms. The arcane-set effect is +0.79 ([+0.17, +1.42], exact $p = 0.032$, Holm-adjusted $p = 0.097$, $r = +0.670$); the thin-set estimate is +0.30 ([−0.03, +0.63]), positive but non-significant. For general questions, +0.05 [0.00, +0.13] lies within the pre-specified ± 0.25 margin, a non-regression reading by interval inclusion (§10.8); its exact p is 0.5 on only two non-zero pairs, and with 58 of 60 pairs tied (mean scores 5.0 loom, 4.95 raw) the instrument is near its ceiling and insensitive on this set. The three set-level point estimates (+0.79, +0.30, +0.05) fall in curation-depth order, but their intervals overlap, so the gradient is three point estimates with overlapping intervals, suggestive rather than established; no interaction test separates a curation-depth effect from other differences between the sets.

Injection engagement is 95.8% on arcane, 100% on thin and 78.3% on general. Thus the general-set null cannot be explained chiefly by the gate skipping ques-

tions, although it skipped the remaining 21.7%. Median injected context is $\approx 1,200$ –1,300 tokens across sets. Median latency is comparable for arcane, 89.8 s (loom) versus 93.4 s (raw), and general, 66.0 versus 65.7 s. Thin is slower under loom, 172.4 versus 145.2 s, consistent with full engagement and longer scaffolds.

10.6 Judge-family robustness

Study 2’s headline is graded by a single gpt-4.1 judge, so a residual concern is that it tracks a family-specific grading style rather than answer content. The same 114 paired answers were therefore re-judged from scratch by claude-opus-4-6, a family used nowhere else in the paper, under the identical rubric, blinding and resume protocol. The contrast reproduces and slightly strengthens: pooled +0.342 (33 non-tied) against +0.272, with the same per-set ordering (arcane +1.042 against +0.792, thin +0.400 against +0.300, general +0.033 against +0.050, still on two to three non-tied pairs). Different provider families limit shared

Table 4: Exposure/recovery decomposition over the 510-question sweep: the same 1,136 gold target instances scored under each of ten models, 11,360 observations pooled rather than 11,360 independent facts. All four counts are given per model. Sorted by context utilisation (Util., $n_{11}/(n_{11}+n_{10})$); n_{11} is exposed gold the answer recovered, n_{10} exposed gold it omitted, n_{01} gold recovered though unexposed and n_{00} unexposed gold not recovered. Each row sums to the 1,136 gold target instances. Across all ten models n_{01} never exceeds 1, and is 3 pooled: recovery beyond exposure is empirically negligible. Counts are pooled over gold items (micro-averaged), whereas the headline recall and ceiling average per-question ratios (macro-averaged), so these counts do not reconstruct those numbers directly.

Model	Util.	n_{11}	n_{10}	n_{01}	n_{00}
Gemini 3.7 Flash	0.955	1012	48	0	76
Gemini 3.5 Flash-Lite	0.947	1004	56	0	76
Claude Haiku 4.5	0.946	1003	57	1	75
GLM-4.6	0.945	1002	58	0	76
Llama-3.3-70B	0.934	990	70	0	76
DeepSeek-Chat	0.931	987	73	0	76
Qwen-2.5-72B	0.924	979	81	1	75
Mistral-Small-24B	0.917	972	88	1	75
GPT-4.1-mini	0.908	962	98	0	76
Gemini 2.5 Flash-Lite	0.900	954	106	0	76
Pooled	0.931	9865	735	3	757

grading style; they do not establish statistical independence.

10.7 Negative controls: attributing the gain

Four controls test whether the live lift reflects specific knowledge transfer or merely the presence of a fluent, on-corpus block. Each holds model and question fixed while perturbing only the scaffold, fetched verbatim from the LLM-free `/loom/scaffold` endpoint (§5.4). The cohort is 57 frozen questions, 24 arcane and 33 thin, answered by the local Qwen3.8-27B through *true*, *shuffled*, *masked* and *irrelevant* at a 1536-token budget, with the *loom* and *raw* arms from the same day’s paired run. Its completions are released. The scores reported here come from re-judging those completions with `openai/gpt-4.1` at temperature 0 under the paper’s rubric verbatim, the same judge family as Study 2’s headline; that re-judge agrees with the archived scores on 227 of 240 rows exactly and on 240 of 240 within one point of the 0–5 scale.

Cohort accounting. The lower panel of Table 6 gives the per-arm flow the varying n rests on. One question, `arc_sov09`, never reached the model in any control arm: the scaffold endpoint declined to inject for it, so the harness wrote a placeholder row rather than a completion. Planned is therefore 57 per arm and attempted 56 in the four control arms, while the *loom* and *raw* arms attempted all 57 because the live harness does not gate on scaffold engagement. The remaining loss is empty output: at 1536 tokens the reasoning backend emits a reasoning block before its answer, and

on 35.7–50.0% of control-arm calls the reasoning alone consumed the budget (§14). Every contrast is paired and computed on its own pairwise-complete question set, which is why the per-contrast n varies and why a direct contrast need not equal the difference of the two corresponding arm–raw rows.

What this cohort supports. Holm–Bonferroni across the whole nine-contrast family leaves one contrast significant: **loom** – **raw** = +0.59 [+0.24, +0.94], Holm p = 0.012. The served path beats no context. **True** – **raw** = +0.59 [+0.13, +1.13] has the same magnitude and an exact p of 0.040, but does not survive correction across the family (p = 0.318). No content-specificity contrast is close: **true** – **irrelevant** is +0.39 [+0.00, +0.87], **true** – **shuffled** +0.30 [−0.10, +0.73] and **true** – **masked** +0.11 [−0.22, +0.48], all with intervals touching or crossing zero and none surviving Holm. A common-intersection sensitivity analysis addresses the objection that different rows rest on different questions: repeating the whole family on the 20 questions graded in every one of the six arms leaves *nothing* significant under Holm, while every point estimate keeps its sign and rough magnitude (loom–raw +0.80, true–raw +0.60, true–irrelevant +0.40). That check establishes consistency within the surviving subset, and nothing wider. The supported claim is therefore narrow. *The served path beats no context on this cohort; whether the scaffold’s specific content matters beyond a well-formed on-corpus block is not established at this power.* Attrition does two things here and only one of them is power. It costs precision, concentrated in exactly the arms the content contrasts need; and since a longer or harder prompt is what exhausts the budget, difficulty and loss may be related, so the surviving rows may describe a different population of questions from the cohort as planned. A service-level effectiveness claim would count those failures rather than condition on an answer being produced.

Wrapper difference, and which contrasts it touches. Placement is *not* the difference between the live path and the controls: the served node and the control harness both merge the block into a system message, appending to the first system message or inserting one at position zero. The verified difference is the wrapper text. The live path always prepends an authority instruction (“treat it as ground truth for definitions and relationships... where it is not relevant, ignore it”), whereas the control arms fetch the bare block from `/loom/scaffold`, which returns no preamble. Because *all* control arms share that weaker wrapper, the contrasts among them (true against shuffled, masked or irrelevant) compare content within one common instruction envelope and are not confounded by it. The contrast that is confounded is **loom** – **true**, which mixes the serving path with the missing preamble; it is reported for completeness and carries no attribution.

Table 5: Study 2, the production-node paired study: Qwen3.8-27B, model held constant, loom versus raw serving path, judged 0–5 by a cross-family GPT judge. Paired mean difference with seeded 10,000-resample bootstrap 95% CI, two-sided exact conditional signed-rank p (Holm-adjusted across the three sets), the matched-pairs rank-biserial r (paired effect size, conditioned on non-tied pairs), and win/loss/tie counts. Complete-case after budget-exhaustion re-runs (§10).

Set	n	Δ (loom–raw)	95% CI	p (Holm)	r	W/L/T
arcane	24	+0.79	[+0.17, +1.42]	0.032 (0.097)	+0.670	10/3/11
thin	30	+0.30	[−0.03, +0.63]	0.130 (0.261)	+0.431	12/5/13
general	60	+0.05	[0.00, +0.13]	0.500 (0.500)	+1.000 [†]	2/0/58
pooled	114	+0.27	[+0.11, +0.45]	0.0023	+0.589	24/8/82

[†] The general-set r rests on 2 non-tied pairs (58 ties); read it as directional only, not as a magnitude.

What the placebo guarantees. Enforcing seed-IRI disjointness, each donor sharing neither a seed IRI nor a seed class with the target gold, is what removes the donor-vocabulary leakage a naive shift-by-one derangement carries. That is a control-design lesson independent of any effect size. But seed disjointness is a *construction guarantee*, not answer disjointness: disjoint seed classes can still share ancestors, relation targets and synonyms, so a donor block may expose some of the target’s gold. What each placebo exposed is not measurable from these rows: the injected bytes are not persisted, and the cohorts carry free-text reference answers rather than the {slug, title} gold items this paper’s matcher needs.

Scope. These controls are a single local model judged on a 0–5 rubric, whereas the ten-model sweep of §10.3 is a lexically scored study on a different cohort. They are different studies; neither inherits the other’s interpretation.

10.8 Out-of-domain: the gate holds

In the five-model judged arm (§9), a cross-family judge scored 60 out-of-domain questions 0–5 against frontier gold. Every harness–bare 95% interval includes zero: all questions −0.05 [−0.11, 0.00], off-domain −0.09 [−0.25, +0.02], and the pre-specified worst case of erroneous injection, off-domain questions on which the confidence gate mis-fired and injected anyway, −0.23 [−0.63, +0.05] on 40 pairs. The adjacent and in-domain-general subsets are +0.01 [−0.02, +0.05] and −0.06 [−0.14, 0.00] on 100 pairs each. The gate correctly skipped 60 of 100 off-domain gradings, making harnessed and bare answers identical; detected degradation was confined to over-firing and concentrated on the weakest model. Study 2 provides the interval-inclusion result: general-set $\Delta = +0.05$ [0.00, +0.13] lies within the pre-specified ± 0.25 margin. That single-model result does not generalise to the five-model arm, whose worst-case subgroup interval [−0.63, +0.05] extends beyond the margin, so equivalence is *not* established there, only not-detected regression. Non-detection under a gate that engaged on 78.3% of general questions is also evidence about the deployed policy rather than proof that the gate causes the non-regression, since there is no matched always-inject ablation.

11 What the Accounting Attributes

Both studies place the graph-grounding uplift over this curated corpus almost entirely within *exposure*: the measured recall is achievable by restating curated facts already shown to the model, so the evaluation cannot attribute it to anything further. That is a statement about what the instrument can credit, not about what the models did. One scope condition holds throughout: the ≈ 0.92 grounded recall is for questions phrased in the graph’s title vocabulary, and under paraphrase the exposure ceiling itself falls to 0.328 (§8), so the headline figure is the high end of a distribution rather than a single operating point.

Exposure-normalised recall is the headline.

Raw uplift is large and consistent across the sweep, while gain over copy is small and uniformly negative: every model appears transformed against its bare-model baseline, none exceeds a verbatim copy of its context, and they differ chiefly in how much exposed gold they drop. Uplift therefore carries little information about the model, while gain over copy places the same ordering on an exposure-normalised scale. Its contribution is the reference point rather than a new ranking, and that is the case for reporting the accounting as standard practice.

Controls attribute less than the uplift suggests.

The served path beats no context, and no content-specificity contrast survives Holm correction (§10.7): true–irrelevant is +0.39 with an interval touching zero, and true–shuffled and true–masked are smaller still. The graded facts are therefore not shown to carry the effect beyond what a well-formed, on-corpus block supplies, which is the shape of the noise confound Cucenasu, Trappolini, Siciliano, et al. [7] warn of. This is non-establishment and not absence of a content effect: the cohort lost 36–50% of its control-arm completions to budget exhaustion and is underpowered for everything except loom against raw. Seed-IRI disjointness still matters as method, because an unconstructed placebo carries donor vocabulary and produces a misleadingly favourable baseline, but it is a construction guarantee rather than answer disjointness, and the irrelevant arm is not length-matched to the true arm.

Table 6: *Upper panel:* negative-control contrasts, pooled arcane+thin, paired mean difference with a seeded 10,000-resample bootstrap 95% CI and Holm-Bonferroni correction across the whole nine-contrast family. Scores are `openai/gpt-4.1` via OpenRouter at temperature 0 under the paper’s rubric verbatim, re-judged on 2026-09-21 from the released completions; the re-judge agrees with the scores archived on 2026-08-17 at 94.6% exact and 100% within one point. [†] is the one contrast surviving Holm. Each contrast is computed on its own pairwise-complete question set, so a direct contrast need not equal the difference of the two arm—raw rows. *Lower panel:* the per-arm flow behind those n . `arc_sov09` is structurally excluded from the four control arms because the scaffold endpoint declined to inject for it; the remaining loss is empty output at the 1536-token budget (§14), and *graded* equals *completed* because every non-empty completion was judged.

Contrast	n	Δ (0–5 scale) [95% CI]	Holm p
loom – raw	54	+0.59 [+0.24, +0.94] [†]	0.012
true – raw	32	+0.59 [+0.13, +1.13]	0.318
masked – raw	33	+0.52 [+0.03, +1.03]	0.417
shuffled – raw	36	+0.25 [−0.36, +0.89]	1.000
irrelevant – raw	28	−0.04 [−0.61, +0.46]	1.000
true – irrelevant	23	+0.39 [+0.00, +0.87]	0.656
true – shuffled	30	+0.30 [−0.10, +0.73]	1.000
true – masked	27	+0.11 [−0.22, +0.48]	1.000
loom – true	32	+0.06 [−0.28, +0.38]	1.000

Arm	Planned	Attempted	Completed	Empty	Empty %
loom	57	57	56	1	1.8
raw	57	57	55	2	3.5
true	57	56	32	24	42.9
shuffled	57	56	36	20	35.7
masked	57	56	33	23	41.1
irrelevant	57	56	28	28	50.0

What the accounting does not price: controlled delivery. The reason an organisation builds a node of this kind is not that a model reasons better with a graph attached to it. It is that the organisation’s own maintained knowledge should be answerable through a model it can replace, with control over which source is consulted, which version of that source is current, and by what route a new assertion becomes authoritative. Selecting that material, keeping it current, governing what may be added to it and delivering the right part of it to a given request are substantial system responsibilities, and most of the engineering in this stack is spent on them (§3.1, §13). Exposure accounting identifies which part of an observed result those responsibilities already explain. It does not measure what they are worth to operate, and a negative gain over copy is not an argument against them: a system can be useful because it reliably makes maintained knowledge available through a model, whether or not its recall exceeds what a copy of the shown material would score. Three statements can be made confidently on this evidence: versioned curated material is served through a replaceable model interface (§4); the scaffold supplies the gold target names for the large majority of in-domain questions (§10.1); and each answer carries source metadata naming the generation it was served from. Three others still need evidence: that this path is cheaper than an alternative at matched quality; that ontology structure outperforms simpler retrieval over the same corpus at an equal budget (§16); and that the attribution metadata is accurate at the level of an individual answer (§15).

Consequence 1: compute the ceiling before you architect. Calculate the ceiling on representative questions before selecting an architecture. A ceiling close to 1 (0.964 here) says answer entity names are exposed in the context. It does not say the required relation is present or that the answer needed no reasoning, so what follows is a prioritisation heuristic and not a diagnosis. Where the ceiling is high, retrieval already supplies most target coverage, so the exposure reference should be reported alongside the quality and system baselines a generative component is judged on; positive gain over copy is not a prerequisite for useful generation, and a large bare-model uplift is exactly what exposure alone predicts. Where the ceiling is low, GraphRAG-style machinery (community summaries, multi-hop traversal, GNN soft-prompts) has room to earn its keep, provided the low ceiling is not simply retrieval or vocabulary failure (§8). We suspect much reported graph-RAG gain is measured in the first setting without the subtraction; the ceiling is cheap enough to compute before expensive architecture is built.

Consequence 2: the instrument doubles as a promotion gate. Where the model’s measured recall is achievable from what the corpus supplies, it is natural to hypothesise that investment should favour upstream curation, whose cost is amortised rather than repaid per query. We state that as a hypothesis only, since this paper measures neither curation effort against query volume nor a quality-matched alternative serving path. What the instrument does support

is narrower. Automated extraction candidates, such as blocks proposed by OntoCast [18] (treated here solely as a third-party upstream producer), can be evaluated through the copy ceiling of the questions they answer. A block that drives that ceiling towards 1 is *name-complete under the matcher*: the answer strings for those questions are present in it. That reading is weaker than completeness of the answers themselves, because the audit of §10.2 shows name presence and asserted relation coming apart on roughly one credited item in fifty. Note the direction: the gate is name-completeness of the source, not positive gain over it, since requiring an extraction to *exceed* what its own source exposes would penalise faithful extraction. The gate is necessary and insufficient: term-stuffing answer names inflates the ceiling while leaving a block incoherent or wrong, so exposure coverage must complement existing CI, consistency and human-review gates.

12 A Production Case Study: Podcast Assertion Extraction

A production deployment tested whether scaffold grounding helps a model *write into* the corpus. The node’s podcast-ingest pipeline⁵ extracts tiered, source-attributed assertions from episode transcripts, each carrying `ontology_terms` used to locate a destination page; assertions whose terms resolve to no page are discarded or queued for page proposal, which makes *term resolution against the live graph* the landing metric. We report a lexical proxy, normalised string matches against the 8,142-class inventory of this generation snapshot, and it is not a human judgement of assertion quality.

We ran the unmodified production prompt over ten episodes of a technology news podcast (November 2025–August 2026) across ten arms: the node’s local Qwen3.8-27B *through the Loom* (scaffold injected at the default 1,500-token budget, verbatim short-circuit disabled) and *bare*; and eight frontier or near-frontier cloud models, all bare, because the Loom grounds only the model behind its own façade. All arms used direct chat-completion calls with identical sampling (temperature 0.2, 12,288 max tokens). Claude and GPT used a model router; the others used their providers’ APIs. One Loom call produced three assertions for one episode and remains in the totals.

Holding weights, prompt and GPU fixed, the paired scaffold effect is large: term resolution rises from 0.17 to 0.55 (3.2×) while assertion count falls, and generation time roughly halves (173s against 346s per episode), consistent with the scaffold displacing open-ended reasoning. Grounding narrows the model’s vocabulary towards the graph’s, as integration requires. Grounded,

the local 27B exceeds every bare frontier arm (best 0.42), reproducing Study 2’s read-path pattern in a write path. This does not establish that frontier models are worse extractors: several produce more assertions, and assertion quality is not judged here. Without the graph’s vocabulary, however, their concept labels miss it and the assertions do not land. Fidelity to noisy input matters too: several arms reproduce speech-recognition errors as entity names (7/181 for GLM-4.7, 3 each for DeepSeek and Opus 4.8), whereas the Loom-grounded Qwen arm emitted none and the bare Qwen arm one, and corrupted names damage page matching and downstream dedup fingerprints. This remains *single-configuration estimation*: one podcast, ten episodes, one run per arm and one lexical scorer give a point estimate from one draw, with no sampling variance and no transfer test. The paired Loom-versus-bare contrast is least exposed because both arms share the draw. This is evidence about where extracted knowledge lands, not a model-quality leaderboard.

12.1 Judged page: an end-to-end quality gate

Term resolution does not show whether a page improves. For each arm, we therefore applied its matched assertions to sandbox copies of target pages using the production integration mechanic. A cross-family judge (Gemini 3.1 Pro, temperature 0, blind A/B with seeded order randomisation, un-blinded at analysis) scored each before/after pair. A second judge, GPT-5.6, re-scored a 20% subset with 93% verdict agreement. Every arm degrades judged page quality (means -1.04 to -0.44); the two arms adjacent to a judge’s family are flagged in the released analysis.

Insertion mechanics alone do not explain the result. A section-splice rewriter emits only an anchored section edit, which deterministic code applies fail-closed while preserving the original by construction, and prompt optimisation over four variants on a dev split, validated on held-out pages under two rubrics (the second written post-hoc, to credit informativeness explicitly), left the best variant negative under both judges (-0.40 dev, -0.90 held-out). Persistence across mechanic, rubric and judge suggests a structural problem: narrow, newly extracted facts dilute a mature curated page.

Under a pre-declared decision rule, the integration phase was disabled while extraction and verification continue, and redesign addresses *placement* rather than phrasing. The blind before/after judged comparison identifies a value-destroying step before it runs at scale; the copy-ceiling name-completeness gate of §11 does not, and cannot, because it grades whether a source exposes the answers to the questions it should answer, not whether inserting that source into a mature page improves the page. The two are complementary halves of one pre-filter (§13), and this case exercises only the judged half.

What the judged gate can and cannot rule out. Contextual judging is itself error-prone [52], so the

⁵The ingest pipeline and the ledger writer of §13 ship in the VisionFlow agent environment: <https://github.com/DreamLab-AI/VisionFlow>; the graph and its CI-gated builder are the knowledgeGraph repository already cited.

Table 7: Assertion extraction across ten serving arms, ten episodes. *Ground* is the fraction of emitted `ontology_terms` that resolve to an existing knowledge-graph page (lexical proxy); *Artif.* counts claims that reproduce speech-recognition artefacts as entity names (e.g. “Opus 48”, “GPT55”). *n* is total assertions; *Conf.* is the model’s own mean stated confidence, which is self-reported and not comparable as a quality measure across families.

Arm	<i>n</i>	Conf.	Ground	Artif.	s/episode
Loom + Qwen3.8-27B (local)	125	0.83	0.55	0	173
Qwen3.8-27B bare (local)	170	0.81	0.17	1	346
Gemini 3 Flash (preview)	107	0.80	0.42	1	13
GLM-4.7	181	0.80	0.39	7	17
DeepSeek v4 Flash	131	0.86	0.28	3	66
Claude Opus 4.8	171	0.75	0.27	3	51
Claude Haiku 4.5	295	0.82	0.22	0	242
GPT-5.6	133	0.82	0.21	0	23
GLM-5.3	169	0.82	0.21	0	33
Claude Sonnet 5	175	0.76	0.20	0	50

degradation is measured with an imperfect instrument, but the most relevant bias would mask rather than manufacture it, since judges favour text labelled as refined [54] and our pairs are blind with seeded order randomisation. The dilution effect, whereby non-diagnostic detail weakens a judgement built on diagnostic evidence [40], is consistent with narrow extracted facts eroding a mature curated page, though style, placement, length and truth of the inserted content are not separated here.

13 Write-Path Lifecycle

In place of the refused integration phase (§12.1), the write path is built from this paper’s two instruments. Extracted, verified assertions land on per-episode *ledger pages* inside the graph, written by the same VisionFlow ingest pipeline that produced the case study above, and wikilinked to the topics they concern: readers see them immediately through linked references, curated prose is never edited, and, because ledger pages carry no ontology markup, nothing enters the served corpus by default. Landing is thereby separated from promotion. A topic whose ledger accumulates evidence, or a thin page whose additions have been judged to improve it, becomes a *candidate*; candidates pass an automated pre-filter consisting of the two instruments this paper developed, the blind before/after quality judgment of §12.1 and the copy-ceiling name-completeness gate of §11. Survivors reach the graph’s existing governed proposal queue⁶ as scored dossiers with provenance down to assertion fingerprints, and approval triggers batched section regeneration, which is the point at which content acquires ontology markup and becomes servable. Direct enrichment of thin pages was evaluated as an alternative landing zone and rejected on its judged evidence: its mean stayed nega-

tive across twenty pairs (−0.65 and −0.60 under two rubrics), and the section-splice mechanic could not apply at all on thirteen of them for want of section anchors, leaving the ledger as the single landing zone. Figure 4 draws the path and the open-source component carrying each stage.

Read as architecture, the lifecycle is this paper’s two instruments put to work: the corpus is the asset, every write to it is measured before it is trusted, and the same ceiling that audits the serving layer gates admission to it. The ledger-and-served split follows the Event Sourcing / CQRS materialised-view pattern [13] *at the assertion layer*: the ledger is the append-only source of truth for extracted assertions and the regenerated sections built from it are disposable projections, which is why a diluting projection can be discarded without data loss. Mature curated prose predates the ledger, is retained as a source in its own right and is never rebuilt from it, so the pattern describes the assertion layer and not the whole corpus. Promotion is idempotent by assertion fingerprint, so the at-least-once delivery inherent to such pipelines cannot double-promote. Nothing in this section is a measurement: it is the architecture the two measured gates were used to choose.

14 Reasoning-Budget Exhaustion

Serving a reasoning model behind an injected scaffold introduces a measurement hazard that conditions several reporting deviations (§10) and constrains the token budgets used throughout. At `max_tokens=1536`, 42 of 234 loom/raw completions in Study 2 exhausted the budget *during reasoning* and returned no text; the four negative-control arms, left at 1536, lost 36–50% of completions the same way (irrelevant 50.0%, true 42.9%, masked 41.1%, shuffled 35.7%; the ordering does not track scaffold disorder, and no per-length analysis was run, so we report the rates without a monotonic claim). The mechanism is a prompt × reasoning-budget inter-

⁶The governed propose/approve write path is the ontology-bridge component of the same stack; the serving façade whose scaffold and copy-ceiling endpoints the pre-filter reuses is the Loom repository already cited.

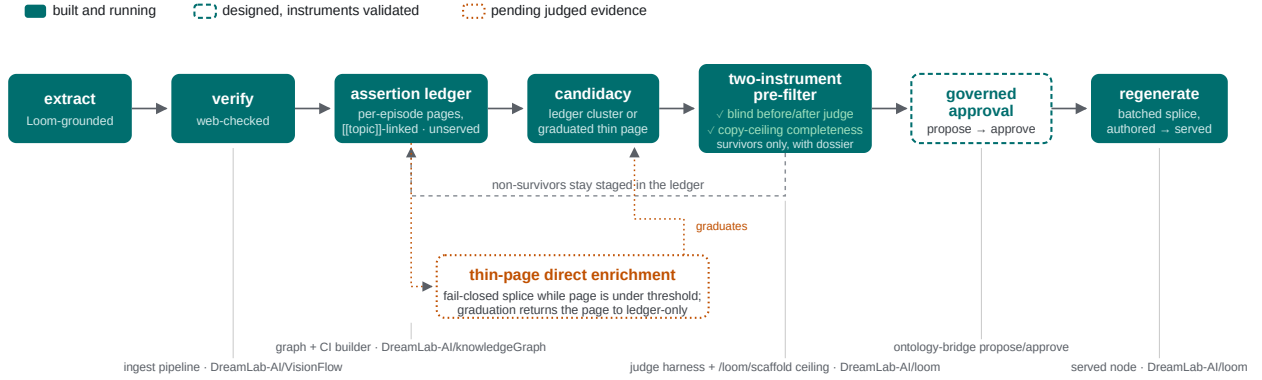


Figure 4: Write-path lifecycle and the open-source component carrying each stage. Landing is separated from serving: assertions stage on unserved ledger pages, candidates pass the two instruments this paper develops, and only governed, regenerated content enters the served corpus. The thin-page branch is conditional on its own judged evidence and self-terminates: an enriched page that crosses the thinness threshold returns to ledger-only treatment.

action, with input and output capacity kept distinct: the scaffold consumes the *context window*, while the reasoning trace and the answer consume the configured *completion allowance*; a longer or noisier prompt induces a longer reasoning trace, and a reasoning model that exhausts the allowance mid-trace emits nothing rather than a truncated answer. Two consequences follow. The completion allowance must cover reasoning-plus-answer, not answer alone: extraction in the case study uses 12,288 tokens and the Study 2 re-runs 4096, and any sub-2000-token allowance is unsafe for a reasoning backend behind a non-trivial scaffold. And empty completions must be detected explicitly rather than scored as wrong, because the first-attempt empty rate is itself a deployment failure signal (42 of 234, 17.9%) and complete-case results measure answer quality conditional on an answer being produced, not service effectiveness. Temperature 0 does not guarantee deterministic remote inference: “deterministic” here refers to scoring, scaffold generation and fixed decoding settings, not to provider-side reproducibility.

15 Open Axes

This node’s deployment bar (§3) resolves into four questions a deployment has to answer about itself. Table 8 states each one, the instrument that bears on it, and its status in this work. Two are measured on limited sets and two are not measured at all. The bar is conjunctive, so evidence on two of the four establishes nothing about the whole.

Questions 1 and 2 are the two this paper instruments, and both statuses are qualified rather than clean: the in-domain figure is recall of exposed gold target names under a sampled relational audit, and the out-of-domain figure is non-detection of regression on one set rather than demonstrated equivalence. Question 3 is the one whose absence most constrains the architectural reading, because a bare model is not the

baseline a curated scaffold competes against. Question 4 is the one that would let “recall of exposed gold” become “attributable”.

16 Companion Work

A companion note⁷ measures a different instrument on a different task: a typed-decision seam in an agent runtime, where a router must choose one of 115 exposed skill rubrics or decline. A choice over shown options has no natural no-op extractor, so its baseline is constructed from a judge-free ranker with an oracle-tuned decline rule. That constructed quantity is ranker-relative: it measures a judge’s advantage over a particular baseline procedure, not the fraction of gold a fixed text exposes. The reusable part is the reporting discipline, that a choice task whose gold sits among its shown options should carry a fair baseline beside its accuracy, tuned to a real optimum and read by population rather than in aggregate. Four lessons from that note bear on any such evaluation: a coarse threshold grid stepped over the ranker’s own optimum and so overstated the judge’s advantage against it; exclusion clauses indexed as positive evidence made turns score highest on the very option written to reject them; aggregate accuracy concealed that the judge’s entire advantage sat on boundary and late-clause turns and was nil on ordinary picking; and systematic ranker–judge disagreement located a labelling error in a corpus its authors had already reviewed.

Architectural consequences of these results, for an organisation holding a large private corpus it must answer from without the content leaving the boundary, would need evidence this paper does not have: a matched comparison of the ontology scaffold against

⁷[docs/research/companion-routing/](#) in the served-node repository, with its corpus, per-run reports, embedding caches, analysis scripts and manifest under [uplift-results/routing/](#). That note states its own release gaps: per-item rows survive for one of its three engines.

Table 8: Deployment questions, the instruments that bear on them, and this work’s status against each. The four are measured on different instruments, so no single number speaks to the conjunction, and no row is closed.

Question	Instrument & evidence	Status
1. Does it recover the corpus’s information in domain?	Lexical exposure accounting and signed gain over copy across ten models (§10.3); stratified relational audit of 423 observations under a cross-family judge (§10.2); paired production quality comparison (§10.5)	Measured on one corpus, one generation and one question set, by a lexical matcher whose credit is audited on a sample rather than in full
2. How does it perform beyond the corpus’s useful coverage?	Judged Δ against frontier gold out of domain and on the general set (§10.8); lexical recovery on gold the scaffold omitted (§10.4)	Equivalence measured by interval inclusion on one general set with a blunt instrument (+0.05 [0.00, +0.13] inside a ± 0.25 margin), and not established in the five-model arm, whose worst-case subgroup interval is wider than the margin; suppression on scaffold-omitted items measured
3. Does it beat competitive retrieval baselines?	The same model and question set grounded in flat-text retrieval over the same corpus, in direct graph query over the closure, and in live web results, each at an equal token budget	Unmeasured. Broadened from a web-search comparison alone: for graph-derived questions the direct graph query is the more demanding of the three (§16)
4. Is its attribution precise, and is the whole answer?	Per-answer generation identity and confidence are already emitted; precision of that metadata, and of the full answer against fabrication, would be scored against the served block	Unmeasured. Source metadata is available on every answer; its accuracy is not assessed, and precision against fabrication is not scored (§17)

a direct graph query and flat-text retrieval at equal budgets, and an accounting of curation effort against query volume.

17 Limitations and Threats to Validity

- **Recall rewards a verbose copier.** The Study-1 metric is recall against graph-derived gold, so a model emitting the entire scaffold verbatim, gold together with arbitrary fabrications, would score at the copy ceiling. Precision against fabrication and the accuracy of the attribution metadata are unmeasured throughout.
- **The audit’s adjudicator is a model.** Every verdict in §10.2 is gpt-4.1’s, not a human’s, and that model is the same family as one of the ten swept models. The quote gate and the deterministic mechanism census bound how far a judge’s own errors can move the estimates, but they do not replace human annotation; the blind sample for that pass is released and unannotated.
- **Single corpus, and dependent questions.** Both studies use one curated ontology in one broad domain, and the magnitudes (a 0.964 ceiling, a +0.27 live lift) do not transfer even though the instrument does. Questions generated from one graph cluster by class, domain and shared relation target, so the domain-clustered bootstrap addresses dependence within this corpus but fifteen domains and one corpus do not establish transfer beyond it.
- **Judge coverage.** Study 2’s headline uses one GPT-family judge, cross-family from the candidate, and a third-family re-judge reproduces the contrast (§10.6); there is no multi-judge panel and no repeatability test across judge seeds.
- **The control cohort is small and the wrapper differs on one contrast.** Per-arm n is 28–56 after 36–50% of control-arm completions were lost to budget exhaustion, and every contrast is computed on its own pairwise-complete question set, so the content contrasts are underpowered and the irrelevant arm is not length-matched to the true arm. The control arms all fetch the bare block from the LLM-free endpoint, so contrasts among them compare content under one common wrapper; only the loom–true contrast is confounded by the live path’s authority preamble.
- **The placebo is seed-disjoint, not answer-disjoint.** Disjoint seed classes can still share ancestors, relation targets and synonyms, so a donor block may expose some of the target’s gold, and the four-arm harness does not persist the injected bytes, so what each placebo exposed is not recoverable from its rows.
- **Semantic fallback disabled throughout.** HNSW semantic retrieval is off because its document-embedding recall (0.816) falls below the

0.87 design floor, so the loom arm is lexical only; §8 measures what that gate is worth.

- **Single production model.** Study 2 fixes the model at Qwen3.8-27B, so its magnitude under other backends is unmeasured; the sweep varies models only on the offline scaffold.
- **Three excluded questions.** Three thin-set questions still empty in one arm at 4096 are excluded pairwise ($n = 114$). The rule is symmetric but the realised missingness was not (loom empty once, raw twice), and a deployment-effectiveness claim would need the empty answers scored as failures.
- **Local-model identity.** The local backend is addressed by a compatibility alias (`qwen3.8-27B`) whose deployed checkpoint is a quantised derivative recorded in the node’s deployment notes rather than an attested reference release; provider display names elsewhere are likewise not immutable version identifiers.
- **Two exposure figures, not interchangeable.** The macro ceiling, 0.964, averages per-question exposed fractions; the item-pooled exposed fraction is $(n_{11} + n_{10})/N = 10,600/11,360 = 0.933$, equivalently 1,060 of 1,136 items per model. Headline recall is macro-averaged and context utilisation is micro-averaged, so the two do not reconstruct each other.
- **One seed, one snapshot.** Questions are sampled under seed 42 and served from one corpus generation. The bootstrap intervals estimate uncertainty under resampling of the frozen questions and do not cover corpus, snapshot, prompt or seed variation.
- **Reproducibility is pinned, but not complete.** Every table and figure resolves to an artefact listed in the manifest at `docs/research/gain-over-copy-paper/MANIFEST.md`, the evidence directories are as of commit `a6d3e41f898c9c18efdf947fc4924ce93c79f4c8` and this manuscript is tagged `paper-v9.2` (<https://github.com/DreamLab-AI/loom/tree/paper-v9.2/docs/research/gain-over-copy-paper>). Two items are absent from that release: the 7.9 MB scaffold index, which is corpus-derived, so a reader can check every derived number against the released rows but cannot regenerate the scaffolds themselves; and two embedding caches, excluded for size and rebuilt by their own released scripts.
- **Equivalence and power.** The out-of-domain reading rests on an instrument near its ceiling: 58 of 60 Study-2 general pairs tie at means of 5.0 and 4.95, so sensitivity to deterioration on harder questions is weak and the equivalence claim is confined to that instrument and set. Power figures quoted here are computed against observed effects and are therefore post hoc.

- **A constructed baseline is a different instrument.** The copy ceiling applies to a task with a natural no-op extractor. For a task without one, such as a discriminative choice over shown options, a baseline must be constructed from a ranker with a decline rule; that quantity is ranker-relative and is not the exposure scalar of §5 (§16).

18 Conclusion

Grounding evaluations whose gold derives from the injected corpus should account for exposure: classify each gold item by whether the shown context exposed it and whether the answer recovered it, report the four counts, and quote the copy ceiling as their per-item scalar. The counts distinguish dropped exposed facts from beyond-exposure recovery, which neither a bare-model baseline nor the scalar alone can do, and depend on nothing but a matcher and fixed text.

What that buys here is narrow and worth stating exactly. Across ten models, gain over copy is uniformly negative and beyond-exposure recovery is three items in 11,360 observations of 1,136 target instances. A large raw-to-grounded uplift therefore establishes delivery of exposed names and nothing beyond it: it does not establish reasoning, and equally does not establish its absence, because zero gain is compatible with successful reasoning over names the context already shows and with offsetting omissions and recoveries that cancel. Exposure accounting and reasoning evaluation are two instruments, and the audit of §10.2 shows where the first one stops. Of the items the matcher credits, 2.1% name the gold target without asserting the relation the question asks for or assert it wrongly. On 60 unexposed and uncredited items the judge finds no hidden correct relational answer, an interval reaching about 6%. And all three credits the matcher issued beyond exposure fail relational adjudication, so credit beyond exposure is rare and, on that census, entirely spurious.

Three further results stand on their own. Rephrasing questions out of the graph’s title vocabulary collapses exposure from 0.964 to 0.328, while the absence-keyed fallback built for that case fires on 2 of 506 questions: a gate keyed to finding nothing cannot detect finding the wrong thing. On gold targets this corpus contains but the retrieved scaffold did not expose, recovery falls from 0.121 bare to 0.004 grounded under this authority instruction and retrieval policy: injection is followed by loss of recall the bare model has. And a paired production study lifts judged quality by +0.27 pooled and +0.79 where curation is deepest, while the negative controls that would attribute that lift to the retrieved content do not: the served path beats no context, and whether the scaffold’s specific content matters beyond a well-formed on-corpus block is not established by those controls: complete-case contrasts describe the surviving questions, and budget-related missingness may select for question difficulty or answer characteristics, so the limitation concerns both power

and potential selection bias.

Cheapness, determinism and judge-freedom are bought at one price: the instrument gives up semantic discrimination, and the audit prices that trade rather than removing it. For a curated corpus the goal was never a model that reasons its way to unseen answers, but one that returns what the corpus was curated to say, with an accounting of how much of its score that explains.

Future work. Four experiments would extend the record here. *(i)* A human annotation pass over the 120-unit blind sample already prepared beside the audit, to place agreement and κ against the model adjudicator. *(ii)* A wrapper ablation on the suppressed items of §10.4: hold the omitted targets fixed, vary the instruction across authoritative, incomplete-context-warning, explicit-supplementation and neutral wrappers, and score by outcome type (omission, contradiction, abstention) rather than right or wrong. *(iii)* A multi-hop design that varies the route while holding the answer name exposed, with entity-preserving edge swaps and direction reversals, which is the instrument the accounting cannot substitute for. *(iv)* A repetition of the negative controls under a common retry policy intended to reduce attrition, with remaining failures reported explicitly, adding a length-matched block of random on-corpus sections as a noise floor [7] and persisting each arm’s injected bytes so the placebo’s own exposure can be measured.

A Judge rubrics (excerpted)

Reproduced here in full, as run, is the production-node semantic answer judge (§9, behind the +0.27 headline), which grades a candidate answer against a reference on a 0–5 scale:

```
You are a strict, fair evaluation judge. Grade the CANDIDATE ANSWER against
the REFERENCE ANSWER for the QUESTION, on a 0-5 integer scale:
5 = fully correct and complete vs the reference; 4 = correct, minor omission;
3 = partially correct, no major error; 2 = mostly wrong or one right point;
1 = on-topic but no correct content; 0 = wrong, irrelevant, or fabricated.
Judge CONTENT match to the reference, not style or verbosity. If the candidate
adds correct information beyond the reference, do not penalise it. If it
contradicts the reference, penalise heavily.
Respond with ONLY a JSON object: {"score": <0-5>, "why": "<one sentence>"}
```

The judged-page gate (§12.1) scores a blind before/after page pair on the same 0–5 shape, returning `factual_grounding`, `relevance`, `coherence`, a `better_version` verdict and a `–2..2 improvement` field scored for version B relative to version A. Rubric A judges quality as a reader would, with no indication of how either version was produced; rubric B is post-hoc and differs only in framing the reader as seeking current, accurate knowledge so that informativeness and currency are weighed alongside prose quality. Rubric B validates rather than replaces the rubric A measurement, and both complete templates with their marked data slots ship with the released harness.

B Worked Example: Question, Gold, Paraphrase and Scaffold

A single item from the sweep makes the copy-ceiling inputs concrete. The question, its gold target set, and the paraphrase are all verbatim from the frozen sets; the paraphrase is this question’s entry in the vocabulary-mismatch stress set (§8), accepted on its first attempt by the gate that requires meaning preservation, no verbatim gold and no seed-title token.

```
QUESTION (id q0003):
  In the DreamLab knowledge graph, what is the immediate parent concept of
  Affective Computing? Also name up to three broader ancestor concepts.
GOLD (slug/title targets asserted by the graph):
  { human-computer-interaction / "Human Computer Interaction" }
PARAPHRASE (stress-set rewrite; tokens "affective", "computing" banned):
  In the DreamLab knowledge graph, what is the direct parent category of
  the field focused on enabling machines to recognize and respond to
  human emotions? Also, list up to three more general ancestor
  categories.
```

The deterministic scaffold served for this question (budget 1500 tokens, max_seeds 4, hops 1, prose off) begins:

The following ontology context was retrieved from a curated knowledge graph. Where it is relevant to the user’s request, treat it as ground truth for definitions and relationships between the concepts it covers. Where it is not relevant, ignore it and answer normally.

```
[ONTOLOGY CONTEXT]
## Affective Computing (ai, maturity: emerging)
Affective computing is a branch of artificial intelligence and human-computer
interaction concerned with systems that can recognise, interpret, process, and
simulate human emotions and affective states. [...curated prose continues...]
is-a: Human Computer Interaction; ancestors: AI Research Area,
      Artificial Intelligence
relations: requires: Emotion Recognition, Annotated Training Data, ...
```

Here the gold title “Human Computer Interaction” appears verbatim in the scaffold (as the `is-a` parent), so the per-item copy ceiling is $c_i = 1$: a no-op extractor of the context would already score the answer. This is the high-exposure case the ceiling reports. Two properties of this frozen item need stating so the example is not over-read. The template asks for broader ancestors as well, but the graph asserts gold here as the immediate parent alone, so the ancestor names visible in the scaffold are context rather than gold. And although the engine ran with `prose off`, each served block always carries its curated definition text: the flag controls additional generated prose, not the block’s own content.

C Two-Edge Composition: a Protocol

This appendix specifies a design, not a result: no quantity below has been measured and no execution is reported. The protocol is recorded because exposure accounting cannot substitute for it: a question can require following two relations while its answer name is already a string in the shown context, in which case exposure is 1, a perfect answer scores 1, and gain over copy is exactly zero (§5).

Question construction. Mine two-edge chains $A \xrightarrow{r_1} B \xrightarrow{r_2} C$ over the restriction-encoded relations of the reasoned closure. A chain is eligible when no direct asserted or inferred $A \rightarrow C$ edge exists, when C does not appear in A 's served block, and when the chain admits at most a small bounded set of valid targets. The question names only A and the two relation types, never B : it asks for the endpoint of a two-edge relation, and the controls below test whether both premises are needed.

Arms. Every context-bearing arm serves the same number of blocks, seed-shuffled, so that the presence or the position of a block does not itself signal relevance. The arms are: *both* premises (A 's and B 's blocks plus seed-disjoint distractors); *b-only* and *a-only* single-premise controls, each padded with distractors to the same block count; *masked*, in which both premises are served with their relation labels replaced by uninformative tokens; *distractor*, in which every block is seed-disjoint from the chain; and *raw*, with no context. The *a-only*, distractor and raw arms bound parametric leakage and placebo response; the *b-only* arm bounds what is obtainable from the answer-bearing block alone.

Scorer and the primary contrast. Score by the same deterministic surface matcher used throughout, against the bounded target set, and report precision alongside recall so that enumeration of the candidate space cannot pass as an answer. Because the *b-only* arm already exposes the target name verbatim, no copy baseline separates it from the both-premises arm; the informative quantity is the contrast $\Delta = \text{both} - \text{b-only}$, which nets out answer exposure. Read each Δ against its own headroom $1 - b_{\text{only}}$: a high single-premise rate caps the detectable difference, so a flat Δ at a high *b-only* rate is uninformative rather than negative evidence.

Decisive controls. Four manipulations, none of which a recall rise on adding a relevant block can replace. *Entity-preserving edge swap*: rewire r_2 so that the correct answer changes while the served names and block lengths are held fixed; a model that follows the changed edge to the changed answer is evidence of composition. *Relation reversal*: reverse the direction of r_2 at matched names and length, so that the same entities admit a different correct target. *Consistent renaming*: replace all chain entities with unfamiliar identifiers consistently throughout the served blocks and the question, which removes parametric recognition while preserving the route. *Bounded precision*: cap the valid target set and score precision, so broad enumeration is penalised rather than credited.

Multiplicity and interpretation. With one Δ per model, apply Holm correction across the panel and report Holm-adjusted p -values, with confidence intervals labelled as unadjusted unless a simultaneous-coverage procedure is defined; without it, boundary assignments between positive, null and negative Δ are expected by chance alone. What the contrasts can identify is the effect of adding route information, and of altering it, under this particular context construction. What they cannot identify is composition ability as a model property: adding A 's block changes distractor competition and answer-selection pressure as well as supplying a missing premise, the masked arm confounds label removal with a change in block surface form, and any grouping of models by the sign of Δ is configuration-specific behaviour rather than a capability ladder.

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